Reinforcement Twinning ~~applied to~~ for Ground-Effect Compensation of a Micro Aerial ~~Vehicles in Ground Effect~~Vehicle Crossing a Finite Obstacle

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***Abstract*—This work presents an implementation of Reinforcement Twinning (RT) applied to a micro aerial vehicle operating in ground effect. The framework couples a Crazyflie 2.1+~~as physical environmentwith its Digital Twin~~ , the physical environment, with a digital twin (DT) whose ~~aerodynamic correction to account for ground effect is progressively identified~~ ground-effect correction is identified progressively from flight data. During a first information phase, the DT and the remaining uncertainty on the corrective parameters guide the design of information-oriented policies to be deployed on the physical environment. During a subsequent performance phase, model-based optimization first provides a ~~safe and computationally efficient~~ DT-screened policy, while a reward-transfer surrogate combines DT and physical environment rewards to guide data-efficient policy refinement despite residual model error. The resulting policy is implemented as a lightweight spatial reference correction that retains the standard onboard feedback controller. Experiments over a finite obstacle show that the proposed workflow constrains the dominant ground-effect response over the tested clearance range and improves trajectory tracking with a limited number of physical interactions. A proof-of-concept extension to different operating conditions further illustrates how the same workflow can adapt a transferred correction when its performance changes away from the nominal condition. These results support the use of an ~~adaptive~~ active DT for model learning and control improvement~~without exhaustive model-free exploration in the physical environment~~.**

***Index Terms*—Reinforcement Twinning, digital twins, micro aerial vehicles, ground effect, system identification, ~~predictive control~~optimal experiment design, Bayesian optimization, Gaussian processes.**

I. Introduction

~~interest and investments on Unmanned Aerial Vehicles (UAVs) hass grown significantly~~

INTEREST and investment in unmanned aerial vehicles have grown in recent years. A recent European Commission outlook places the European commercial airborne-drone market at about EUR 14.5 billion by 2030 and indicates that it could exceed EUR 50 billion by 2033 [1]. ~~Such increase in interest is mainly supported by the intrinsic features of these vehicles, including~~ This growth rests on low operating cost, rapid deployment, and the possibility of ~~substituting or support~~ replacing or assisting human operators in hazardous missions. ~~Their applications now span different domains,~~ Applications range from logistics to emergency response and infrastructure inspection. ~~In specific applications, drones of reduced size can carry important advantages. These includes especially operations~~ Small vehicles are particularly useful in cluttered environments ~~, where accessibility is limited and dangerous for human operators. In such context, the size of Micro Aerial Vehicles~~ that are difficult or dangerous for people to enter. There, micro aerial vehicles (MAVs) ~~plays a major role as they can provide a preliminary mapping of the restricted zone with small onboard sensor to guide deployment of human operations if retained necessary. Indoor~~ with small onboard sensors can map a restricted zone before human operators are sent in.

In indoor inspection, monitoring, and ~~emergency-response missions are representative examples. A~~ emergency response, a MAV can enter confined spaces and cross passages whose clearance is only a few rotor radii. It can approach machinery, beams, pipes, and damaged structures ~~that would otherwise require~~ without shutting down a facility or exposing an operator to danger. These missions are also ~~among the most~~ demanding. Satellite positioning is generally unavailable, onboard sensing and computation are limited, and ~~communication can be degraded by~~ the surrounding structure ~~. At the same time, obstacle avoidance leaves~~ can degrade communication. Nearby surfaces leave little room for tracking errors ~~. The boundaries therefore become part of the aerodynamic environment rather than passive geometric constraints: the wakes generated by the propellers interact with nearby~~ and also change the flight dynamics. The propeller wakes interact with these surfaces and alter the forces and moments acting on the vehicle, so the surfaces are part of the aerodynamic environment rather than passive geometric constraints.

~~The elementary interactions with nearby surfaces~~ Rotor–surface interactions are usually classified as ground, wall, and ceiling effects according to the orientation of the surface relative to the rotor plane [2], [3], [4]. In a cluttered environment they seldom appear in isolation. A vehicle passing through a narrow tunnel may experience floor, ceiling, and lateral-wall interactions at the same time,

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with an asymmetric flow field that affects both translation and attitude. Even an isolated MAV rotor responds differently in a wall–ground corner than above a floor [5]. Wall and ceiling effects have received less attention than ground effect, and most available studies concern hovering rotors or vehicles. Wall effect ~~is generally~~ remains weak until the rotor is ~~very~~ close to the surface~~; ~~. On a Crazyflie, Carter et al. [3] ~~, for example, measured a substantially smaller change in lift~~ measured a lift change below 5% near a lateral wall~~than ~~, against up to about 20% near a floor ~~or~~ and 60% near a ceiling. Ceiling effect is also ~~localized at~~ confined to small clearances, ~~although~~ where wake blockage and recirculation ~~can produce a pronounced force increase there~~increase the force [4], [6].

Ground effect (GE), by contrast, has been ~~investigated since the middle of the twentieth century in classical aviation and rotorcraft. An underlying surface changes the pressure field and redirects the rotor wake radially , reducing~~ studied since the 1950s for helicopter rotors. A surface below the rotor redirects the wake radially and reduces the induced velocity and ~~induced~~ the power required for a given ~~lift. This can improve aerodynamic efficiency during takeoff, hover, and low-altitude flight. Classical momentum-theory relations, including the Cheeseman-Bennett approximation, describe the effect for~~ thrust. Cheeseman and Bennett derived the resulting thrust gain of an isolated rotor ~~hovering above an extended plane primarily through its normalized clearance. At MAV scale, however, four nearby rotors interact with one another and with~~ from its clearance-to-radius ratio and forward speed [7], and rotor experiments showed that the near-ground flow changes regime with advance ratio and depends on translational acceleration [8]. In multirotors, rotor–rotor interaction and the flow beneath the central body ~~, while low Reynolds number and wake recirculation make the response vehicle dependent [2], [3] . Forward motion adds wake convection and a finite residence time over the surface. Kan~~ change this response [2] . It has been described by adding image rotors and an empirical body coefficient to the classical model [9], or by exponential, quasi-steady, and Gaussian-process models of the measured thrust [10], [11], [12]. On a tethered Crazyflie 2.0, the near-ground lift increase exceeded twice the classical prediction [3], whereas experiments and computations on other quadrotors report a nonmonotonic thrust or power response near one to one and a half rotor radii [13], [14].

Flight near obstacles adds two complications. First, the surface may lie under only some of the rotors. The resulting thrust imbalance produces a moment that a cascaded PID controller rejects poorly [9]. For a single rotor, the thrust under partial coverage has been correlated with the overlap area between rotor disk and surface [15]. Free-flight tests near a building-like edge, however, found that full GE develops only after most of the vehicle has crossed the edge [16]. Second, GE is not static. It varies with forward and climb speed [17], [18] and with the motion of the surface [19], and dynamic quadrotor tests have complemented the static characterizations [20].

Control studies compensate GE online or learn it from flight data. Adaptive control [21] and disturbance observers [10], [22] counter the near-ground force, and Li et al. ~~[17] confirmed the expected efficiency gain for small multirotors in forward flight , but also showed that clearance alone is insufficient and that velocity changes the response~~[22] also fitted a GE model to observer estimates recorded in hovering flights. Deep networks trained on flight data [23] and richer models that include surface motion and past states [19] or GE torques and rotor drag [24] provide feedforward or model-based compensation. In aircraft, GE coefficients have been identified from flight data recorded without dedicated maneuvers [25]. Flight-vehicle system identification, by contrast, designs the inputs to maximize the information on the unknown parameters [26], [27], [28] . Robot trajectories have been synthesized to maximize the Fisher information [29], and informative trajectories have been planned for other multirotor parameters [30]. To our knowledge, no study has selected the flights of a multirotor by their information on the GE parameters.

~~These limitations matter~~ Accurate GE prediction matters in cluttered flight because ~~the purpose of~~ an aerodynamic model ~~is not only to reproduce a force measurement. The vehicle must remain on a safe path while moving~~ must do more than reproduce a measured force. It must let the controller keep the vehicle on a collision-free path from one obstacle to the next. A thrust ~~variation~~ change above one surface can produce a ~~small~~ vertical excursion that ~~appears benign locally but consumes~~ is harmless locally but uses up the clearance available at a following beam, ceiling, or doorway. ~~Predicting and compensating the interaction before the~~ The interaction must therefore be predicted and compensated for before a tracking error develops~~is therefore a safety requirement . The problem is especially clear over finite obstacles, where the~~ . Finite obstacles make this requirement explicit. The vehicle enters GE, remains in it briefly, and then crosses a trailing edge~~. Leading-edge~~ , so leading-edge exposure, transient wake development, and post-obstacle recovery all occur within one mission.

Most MAVs rely on cascaded proportional-integral-derivative (PID) loops. ~~Their low computational cost and robust nominal behavior make them well suited to~~ They are cheap to compute and robust near their nominal operating point, which suits small onboard processors. However, they are ~~intrinsically reactive: an~~ reactive. An unmodeled disturbance must first produce an error before feedback counters it~~. Close~~ , and close to an obstacle ~~,~~ this delay can consume the available safety margin. ~~Model-based predictive methods can~~ Predictive methods instead modify the command or reference before the disturbance ~~fully~~ develops. Model predictive control ~~(MPC), in particular, offers a systematic way to optimize~~ optimizes future behavior while accounting for dynamics and constraints [31], also on quadrotors with learned residual dynamics [32]. Its effectiveness ~~is nevertheless tied to the model used for prediction and the computational burden related to its online~~computation depends on the prediction model and on the cost of solving the optimization online. An incomplete model can ~~generate a confidently~~ produce a wrong correction, whereas a more faithful aerodynamic model may

be difficult to identify and too expensive to optimize repeatedly on the limited hardware of a micro droneMAV.

Reinforcement Twinning (RT) was introduced to address this tensiontrade-off. It couples a physical environment such as a MAV system, called the physical environment, with an active physics-based Digital Twin digital twin (DT) [33], [34], [35]. A conventional DT is often understood as a virtual representation on which the behavior of a physical environment can be reproduced. An active DT additionally learns from physical observations and uses the updated representation to influence subsequent operation. It acts as both a *steering representation* and an *acting model* in the decision process [36], [37]. Experimental data refine its uncertain physics, while virtual rollouts guide the next action of and virtual rollouts select the next policy deployed in the physical environment. This principle has been explored for RT has been applied to wind-turbine control, gust rejection on a flapping-wing vehicle, and cryogenic thermal management, and [33], to the hybrid control of a flapping-wing flight [33], [35], but still poorly experimentally validateddrone [35], and to a flow-control benchmark [34], all in simulation.

Early RT algorithms combined model-based and model-free control. A model-based policy supplied an efficient, physically informed starting point, while and a model-free policy could improve when the model kept improving when the model-based one reached a local or structurally biased optimum. A referee switched between them the two according to their observed or predicted performance. When learning stagnated, policy cloning transferred copied the weights of the more successful better policy to the other learner, allowing the search to resume so that the search resumed from a better region [33], [35]. The model-based branch thus protected made the first interactions less risky and accelerated early learningwithout forcing , without tying the final controller to remain at the model-based optimum.

More recent RT formulations cast the same exchange exchange between model-based and data-driven learning as a twin-calibrated Bayesian policy search [34]. They retain two purposes for a physical episode , in which each physical episode has one of two purposes. An information-oriented policy is selected because its response can chosen to improve the uncertain physics in the DT; , and a performance-oriented policy is selected because it is expected to improve control. The two objectives are coupled through the accumulated data and the current DT, but they need not be collapsed merged into one scalar objectiveat every interaction. This also . This distinguishes RT from dual control. Dual control balances the immediate , which balances the regulation and information consequences of each action [38], [39], [40]. RT instead gives explicit assigns the two roles to separate episodesand connects them through factual replay and the evolving DT, linked by the DT, which is updated by replaying the recorded episodes (factual replay). Once the information collected from the physical environment is sufficient physical data constrain the DT well enough for the task, the updated DT becomes the main instrument for policy search. It can solve DT solves the model-based control problem offline and evaluate candidate policies before they are deployedscreens candidate policies against state and clearance constraints before deployment. The resulting model-based policy provides a physically interpretable anchor: it already accounts for the dynamics learned from the experimentsand can be screened against state, clearance, and actuation constraints. If the final policy . If it is expressed as a compact feedforward correction or reference governor, only this lightweight representation must run onboard, while reference correction, the standard feedback controller remains responsible for stabilization. is left unchanged and still stabilizes the vehicle.

The role of Bayesian RT does not end at the model-based policy. Identification reduces parametric uncertainty but cannot remove a the structural discrepancy between a compact physics model and the physical environment. Adding Additional states and correction terms may reduce this model discrepancy, yet an affordable DT will still omit but a DT that remains cheap to optimize still omits part of the physical response. Bayesian RT therefore RT therefore also learns how control performance transfers from the DT to the physical environment. Each tested policy supplies yields a DT reward and a physical reward. An In the resulting reward-transfer surrogate, an affine relation captures the global calibration between the two , while rewards, and a Gaussian process (GP) represents the remaining GP discrepancyand the uncertainty associated with discrepancy, whose uncertainty is largest in sparsely explored regions [41], [42]. The latter accounts for the part of the transfer that is not described consistently by either the physics model or the affine trend. This is the autoregressive multi-fidelity structure of Kennedy and O'Hagan [43], [41], [44], [42].

This reward-transfer The surrogate turns the final stage into a data-efficient last stage of RT into a Bayesian policy search. Expected Improvement balances (EI) weighs the predicted physical reward against the uncertainty of the surrogate surrogate uncertainty and proposes a candidate that is promising, insufficiently explored, or both [45], [44]. After the candidate is tested, its physical reward [45], [46]. The physical reward of the tested candidate updates the surrogate, and the search is repeated. The policy can consequently search can therefore move beyond the model-based policy even if the DT retains a systematic bias. In contrast with a fully Unlike a model-free search, however, exploration begins from an improved physics-based solution and remains it starts from the model-based policy and stays within a neighborhood that has already been the DT has already simulated and screenedby the DT. The model . The DT supplies the initial direction and the safety envelope; admissible set, and physical data correct its ranking where this is needed. Information-oriented and performance-oriented interactions can in principle be interleaved within this loop, with needed.

The repeated mission also invites comparison with two established approaches. Iterative learning control refines a feedforward signal from the tracking errors of previous trials of the purpose of each episode made explicit by the RT selectorsame task [47], and it has been applied to quadrotor trajectory tracking [48], [49]. Bayesian optimization has tuned quadrotor controllers directly on hardware [50] and has combined simulations and physical experiments as

information sources of different cost [51]. RT differs from both in that the flights first identify an interpretable twin, which then designs and screens the policies, and the physical data correct the twin's ranking of candidate policies.

This paper presents an experimental demonstration of ~~the present RT formulation~~ a two-stage form of RT on a physical MAV. A Crazyflie 2.1+ crosses a finite horizontal obstacle at constant forward speed, ~~experiencing the onset and removal of ground effect within the same~~ so that GE appears and disappears within one mission. The ~~starting~~ mission is repeated over an obstacle of known position and geometry, with external positioning, which isolates the GE transient from perception and planning. The initial DT is the ~~independently validated~~ six-degree-of-freedom Crazyflie model of Richardson and Pretorius [52], ~~augmented with a deliberately low-order static ground-effect law whose unknown parameters are initialized from the Cheeseman-Bennett model. Its limited structure retains a realistic discrepancy, particularly around~~ with an added static GE law whose two parameters start from the Cheeseman–Bennett values. This low-order structure can be identified and used for control design, but it leaves a model discrepancy, mainly at the obstacle edges~~, while remaining suitable for identification and control design~~.

~~The campaign first uses~~ In the first stage, E-optimal information-oriented ~~policies to identify these parametersand then freezes the updated model. The DT subsequently produces~~ flights identify the two GE parameters, which are then frozen. In the second stage, the DT computes a model-based policy~~implemented as a lightweight spatial reference governor~~, applied as a spatial reference correction above the unchanged onboard PID controller. Physical rewards ~~from~~ of nearby policies calibrate the reward-transfer surrogate, and ~~Expected Improvement~~ EI searches for further ~~performance gains. Information-oriented and performance-oriented episodes are treated as consecutive phases in this work, making their experimental roles explicitwhile retaining the same RT loop~~gains. The two stages are consecutive rather than interleaved, so that the role of each flight is explicit. Finally, the correction learned at the nominal condition is transferred to two other combinations of clearance and speed, and a short EI search is tested at one of them.

The paper makes three contributions. First, it applies RT to a physical vehicle. A differentiable DT of a Crazyflie 2.1+ and its onboard controller designs both the identification flights and the control reference. Second, E-optimal flights over two clearance domains identify a static GE law whose amplitude is about twice the Cheeseman–Bennett value, and the DT yields a reference correction that raises the physical tracking reward from 0.544 to 0.855 without changing the onboard controller. Third, after nine EI-selected policies, a reward-transfer surrogate finds a correction with physical reward 0.952 that the DT ranks as nearly equivalent to the model-based one. The transfer of this correction to two off-nominal conditions is also tested.

The remainder of the paper is organized as follows. Section II ~~introduces the RT workflow~~describes the RT loop, the physical environment and DT, and the ~~information and performance phases.~~ test campaign; Section III ~~presents and discusses the experimental campaign following the phases defined in the methodology.~~ reports the experiments; and Section IV ~~summarizes the main findings~~ states the conclusions and limitations.

II. Methodology

A. Reinforcement Twinning Framework

~~Reinforcement Twinning (RT) couples the evolution of a physical environment with that of~~ In RT, the physical environment and an active, physics-based ~~Digital Twin (DT) [33], [34]. The two environments~~ DT evolve side by side ~~and pursue complementary objectives. The physical environment has a performance objective related to the policy parameters to be optimized, whereas the DT must improve the physical model that predicts the response of the physical environment. Experience from the physical environment is therefore used both as evidence to~~ with complementary objectives [33], [34]. Physical data update the DT and ~~as feedback on the policies designed with that DT. Conversely, the DT not only adapts from observations but also~~ score the policies it designs, and the DT proposes the next experiments and ~~control policiesfrom which new observations are generated~~policies.

The two environments and their objective: The physical environment is denoted by a filled circle ($\bullet$) and the DT by an empty circle ($\circ$), following the black- and white-box symbology, respectively. A physical interaction is indexed by n, while k denotes the time step within that interaction. The hidden physical state $\mathbf{s}_{\bullet,k}^{(n)}$ evolves according to

$$\mathbf{s}_{\bullet,k+1}^{(n)} = \mathbf{f}_{\bullet}\left(\mathbf{s}_{\bullet,k}^{(n)}, \mathbf{u}_{\bullet,k}^{(n)}, \mathbf{d}_{k}^{(n)}\right), \tag{1}$$

where $\mathbf{u}_{\bullet,k}^{(n)}$ is the command generated under the deployed policy and $\mathbf{d}_{k}^{(n)}$ collects exogenous variables and disturbances that are not chosen by the policy. These may include environmental loads, variations of the available actuation, imperfect initial conditions, or disturbances in the measurement chain. The state is not assumed to be directly available as in a black-box framework. What enters the RT loop is the noisy observation

$$\mathbf{o}_{\bullet,k}^{(n)} = \mathbf{h}_{\bullet}\left(\mathbf{s}_{\bullet,k}^{(n)}\right) + \boldsymbol{\nu}_{k}^{(n)}, \tag{2}$$

with $\boldsymbol{\nu}_{k}^{(n)}$ representing measurement noise and unmodeled observation errors. The complete experimental record also contains the deployed policy, the applied inputs, the relevant operating conditions, and the physical reward $\mathcal{R}_{\bullet}^{(n)}$.

The DT reproduces the same state evolution with a parameterized transition model,

$$\mathbf{s}_{\circ,k+1}^{(n)} = \mathbf{f}_{\circ}\left(\mathbf{s}_{\circ,k}^{(n)}, \mathbf{u}_{\circ,k}^{(n)}, \mathbf{d}_{k}^{(n)}; \boldsymbol{\theta}_{p}^{(m)}\right), \tag{3}$$

and generates the corresponding DT observation

$$\mathbf{o}_{\circ,k}^{(n)} = \mathbf{h}_{\circ}\left(\mathbf{s}_{\circ,k}^{(n)}; \boldsymbol{\theta}_{p}^{(m)}\right), \tag{4}$$

where the vector $\boldsymbol{\theta}_{p}^{(m)}$ contains the unknown or uncertain model parameters to be identified and m counts their updates. Every physical episode increments n, whereas m advances only when model identification is active; therefore, $m \leq n$.

During factual replay, the DT receives the recorded policy, exogenous inputs, operating condition, and a reconstructed initial state. The mismatch between $\mathbf{o}_\circ$ and $\mathbf{o}_\bullet$ then measures how much of the physical response is not explained by the current twin. During virtual evaluation, the same DT predicts the outcome of policies that have not yet been tested physically.

~~As anticipated, each of the two environments involved in the Reinforcement Twinning loop features~~ Each environment has its own objective. The physical environment ~~aims for~~ seeks policy parameters $\boldsymbol{\theta}_\pi$ in the domain Θ_π that maximize the expected physical reward,

$$\boldsymbol{\theta}_\pi^\star = \arg\max_{\boldsymbol{\theta}_\pi \in \Theta_\pi} \mathrm{E}[\mathcal{R}_\bullet(\boldsymbol{\theta}_\pi)]\,, \tag{5}$$

whereas the digital twin searches for model parameters $\boldsymbol{\theta}_p$ in the domain Θ_p that make factual replay consistent with the accumulated observations,

$$\widehat{\boldsymbol{\theta}}_p^{(m+1)} = \arg\min_{\boldsymbol{\theta}_p \in \Theta_p} \mathcal{L}_p(\mathcal{B}_n; \boldsymbol{\theta}_p)\,, \tag{6}$$

where $\mathcal{B}_n = \{\mathcal{E}^{(1)}, \ldots, \mathcal{E}^{(n)}\}$ is the replay buffer of physical episode records $\mathcal{E}^{(n)} = \left\{\boldsymbol{o}_\bullet^{(n)}, \boldsymbol{\theta}_\pi^{(n)}, e_n, \mathbf{d}^{(n)}, \mathcal{R}_\bullet^{(n)}\right\}$, and $\mathcal{L}_p$ compares the DT and physical observations selected for identification. ~~Equations (5) and (6) are coupled: the model determines which policies appear promising or informative, while the resulting physical records determine how the model and the policy search evolve.~~

Policies and interaction schedule: The interaction between the two environments is mediated by control policies that are designed within the DT and subsequently deployed in the physical environment. ~~Policy design is therefore performed virtually: candidate interactions can be evaluated and optimized before the selected policy is transferred to the physical environment.~~ The resulting physical episode then provides the experimental evidence required by the learning process. Depending on the current objective, the DT designs either an information-oriented or a performance-oriented policy. The information-oriented policy $\pi_\iota(\cdot \mid \boldsymbol{\theta}_\iota)$, with $\boldsymbol{\theta}_\iota \in \Theta_\iota$, is designed to generate observations that are informative about the model parameters $\boldsymbol{\theta}_p$. Conversely, the performance-oriented policy $\pi_\bullet(\cdot \mid \boldsymbol{\theta}_\pi)$, with $\boldsymbol{\theta}_\pi \in \Theta_\pi$, is designed using the current DT to maximize the expected physical reward. The two ~~policy families may have significantly different parameterizations because their classification reflects~~ families are defined by the objective used ~~during policy design~~ to design them and may have different parameterizations. However, the data generated by each physical episode can still support both learning objectives. A performance-oriented policy may excite dynamics that improve the model-parameter estimate, while an information-oriented policy also yields a physical reward~~that supports subsequent performance-oriented policy design~~. The present campaign does not use these rewards, because the information-oriented references depart from the nominal clearance. A binary ~~interaction schedule~~ selector $e_n \in \{0, 1\}$ determines which policy is designed and deployed ~~during each~~ in physical episode n~~:~~,

$$\pi_\bullet^{(n)} = \begin{cases} \pi_\iota\left(\cdot \mid \boldsymbol{\theta}_\iota^{(n)}\right), & e_n = 1, \quad \text{information-oriented,} \\ \pi_\bullet\left(\cdot \mid \boldsymbol{\theta}_\pi^{(n)}\right), & e_n = 0, \quad \text{performance-oriented.} \end{cases} \tag{7}$$

In the general RT formulation, these two types of interaction are freely alternated during the campaign as the twin and policy evolve. For simplicity, the present work ~~adopts a sequential schedule: performs~~ all information-oriented ~~missions are performed first~~episodes first, until the model ~~satisfies the adopted identification and convergence requirements~~meets the stopping criterion of the information phase. The model parameters are then frozen, and the campaign proceeds with performance-oriented ~~missions. We consequently distinguish a first~~ episodes. This defines an *information phase* ~~and a subsequent separated~~ followed by a separate *performance phase*~~, which are described in detail below~~.

Information phase: In the information phase, $\boldsymbol{\theta}_p$ is iteratively updated using the accumulated experimental data, after which the updated DT designs the next information-oriented policy. Given the current replay buffer, Eq. (6) is solved until the model update has converged. The updated DT then generates a set of admissible candidate interactions within the information-oriented policy domain. ~~Each candidate represents a possible next experiment and is evaluated in the DT before deployment.~~ The DT predictions along these interactions are differentiated with respect to $\boldsymbol{\theta}_p$. As $\mathbf{S}(\boldsymbol{\theta}_\iota) = \partial \mathbf{o}_\circ / \partial \boldsymbol{\theta}_p$ denotes the sensitivity of the predicted observations and $\boldsymbol{\Sigma}_\nu$ their noise covariance, the information expected after a candidate policy is

$$\mathbf{F}_{n+1}(\boldsymbol{\theta}_\iota) = \mathbf{F}_n + \mathbf{S}^\top(\boldsymbol{\theta}_\iota)\boldsymbol{\Sigma}_\nu^{-1}\mathbf{S}(\boldsymbol{\theta}_\iota), \tag{8}$$

where $\mathbf{F}_n$ is the Fisher information accumulated from the experiments already stored in $\mathcal{B}_n$. The policy deployed in episode $n+1$ is selected with an E-optimal criterion~~,~~ [26], [53], [54]~~,~~

$$\boldsymbol{\theta}_\iota^{(n+1)} = \arg\max_{\boldsymbol{\theta}_\iota \in \Theta_{\iota,\mathrm{adm}}} \lambda_{\min}[\mathbf{F}_{n+1}(\boldsymbol{\theta}_\iota)]\,. \tag{9}$$

Maximizing the smallest eigenvalue targets the least-observed joint parameter direction, so the E-optimal criterion already favors novelty in the information collected. The admissible set ~~$\Theta_{\iota,\mathrm{safe}}$~~ $\Theta_{\iota,\mathrm{adm}}$ is obtained through virtual rollouts and may enforce state, input, rate, and clearance constraints. ~~This selection may be complemented by an explicit novelty criterion that compares each candidate with the existing dataset. It allows a mission with a slightly lower predicted information content to be selected when its response is sufficiently different from the experiments already available.~~

The selected policy is deployed in the physical environment, its record is appended to $\mathcal{B}_n$, and the model is identified again from the enlarged dataset. Two convergence levels are distinguished. Each parameter update is complete only when the identification loss, the model parameters, and the projected gradient remain within prescribed tolerances over a monitoring window. The information phase ~~is terminated when the converged parameter estimates remain stable after adding new~~

~~interactions and~~ ends when new episodes no longer change the converged estimates appreciably or reduce the magnitude of their ~~estimated correlationno longer decreases appreciably. At this point, further experiments yield only marginal changes in the identified model and in the joint identifiability of its parameters~~correlation. In the present campaign this was judged from the parameter history, the correlation, and the replay of the baseline episode (Section III-B); no numerical threshold was set in advance.

The converged model parameters, denoted by $\widehat{\boldsymbol{\theta}}_p$, are then used to compute the model-based policy π_{MB}, ~~accounting for safety constraints and actuator limits within~~ subject to the reference-amplitude and clearance constraints checked in the DT,

$$\boldsymbol{\theta}_{\pi,\mathrm{MB}} = \arg \max_{\boldsymbol{\theta}_\pi \in \Theta_{\pi,\mathrm{adm}}} \mathcal{R}_\circ\left(\boldsymbol{\theta}_\pi; \widehat{\boldsymbol{\theta}}_p\right). \tag{10}$$

In the present work, π_{MB} marks the transition between the two phases. In a more general RT approach, the model-based policy can be updated after every $\boldsymbol{\theta}_p$ update.

Performance phase: In the present work, the performance phase starts from $\boldsymbol{\theta}_{\pi,\mathrm{MB}}$ and keeps $\widehat{\boldsymbol{\theta}}_p$ frozen. The search is restricted to a neighborhood of π_{MB} because this model-based policy ~~already improves~~ is expected to improve upon the baseline policy and provides a physically informed reference. ~~Exploring the full policy space with a purely model-free strategy would require substantially more physical tests and could expose the physical environment to unsafe policies.~~ The objective is therefore to improve upon π_{MB} by learning the difference between the DT reward and the physical reward. A small set of ~~safe~~ policies around π_{MB} is first ~~evaluated~~ screened in the DT ~~to verify safety and then in the physical environment~~and then flown. These paired rewards initialize the reward-transfer surrogate,

$$\begin{aligned} \mathcal{R}_\bullet(\boldsymbol{\xi}) &= \gamma_0 + \gamma_1 \mathcal{R}_\circ(\boldsymbol{\xi}; \widehat{\boldsymbol{\theta}}_p) + \delta(\boldsymbol{\xi}) + \varepsilon, \\ \delta &\sim \mathcal{GP}(0, k), \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2). \end{aligned} \tag{11}$$

The descriptor $\boldsymbol{\xi} = [\boldsymbol{\theta}_\pi^\top, \boldsymbol{\chi}^\top]^\top$ contains the policy parameters and the operating-condition vector $\boldsymbol{\chi}$ that describes the performance task. The intercept γ_0 and slope γ_1 describe the global affine relation between the DT and physical rewards. The ~~GP~~ Gaussian-process (GP) discrepancy $\delta(\boldsymbol{\xi})$ models the policy-dependent part of the transfer that ~~is not explained by that relation~~ the affine relation does not explain. Its Gaussian-process prior assigns similar discrepancies to nearby descriptors through the kernel $k(\boldsymbol{\xi}, \boldsymbol{\xi}')$, while ε represents experimental and reward-evaluation noise. The present campaign reduces σ_ε to a numerical jitter (Section II-C). The GP is conditioned on the ~~discrepancies~~ residuals between the physical rewards $\mathcal{R}_\bullet$ and the corresponding affine predictions obtained from the DT rewards $\mathcal{R}_\circ$ collected in the replay buffer $\mathcal{B}_n$. Its posterior provides the mean GP discrepancy $\mu_\delta(\boldsymbol{\xi})$ and its standard deviation $\sigma_\delta(\boldsymbol{\xi})$. The predicted physical reward therefore has mean

$$\mu_\bullet(\boldsymbol{\xi}) = \gamma_0 + \gamma_1 \mathcal{R}_\circ(\boldsymbol{\xi}; \widehat{\boldsymbol{\theta}}_p) + \mu_\delta(\boldsymbol{\xi}) \tag{12}$$

and GP interpolation uncertainty $\sigma_\bullet(\boldsymbol{\xi}) = \sigma_\delta(\boldsymbol{\xi})$. ~~This quantity is the posterior uncertainty of the adopted surrogate, not a confidence interval for experimental repeatability. The DT reward supplies the global performance trend, while the GP learns the local discrepancy between the DT and physical rewards.~~

The next performance-oriented policy is selected through Expected Improvement. Let $\mathcal{R}_\bullet^+$ denote the best physical reward ~~and $\xi \geq 0$~~ observed so far and $\zeta \geq 0$ an optional exploration margin. For $\sigma_\bullet(\boldsymbol{\xi}) > 0$, define ~~$Z(\boldsymbol{\xi}) = [\mu_\bullet(\boldsymbol{\xi}) - \mathcal{R}_\bullet^+ - \xi]/\sigma_\bullet(\boldsymbol{\xi})$~~ $Z(\boldsymbol{\xi}) = [\mu_\bullet(\boldsymbol{\xi}) - \mathcal{R}_\bullet^+ - \zeta]/\sigma_\bullet(\boldsymbol{\xi})$, so that

$$\mathrm{EI}(\boldsymbol{\xi}) = \left[\mu_\bullet(\boldsymbol{\xi}) - \mathcal{R}_\bullet^+ - \zeta\right] \Phi[Z(\boldsymbol{\xi})] + \sigma_\bullet(\boldsymbol{\xi}) \phi[Z(\boldsymbol{\xi})], \tag{13}$$

where Φ and ϕ are the standard-normal cumulative distribution and density functions. EI is set to zero when $\sigma_\bullet(\boldsymbol{\xi}) = 0$. Its first term favors policies expected to exceed the best physical reward, while the second favors uncertain candidates. The next interaction is

$$\boldsymbol{\xi}^{(n+1)} = \arg \max_{\boldsymbol{\xi} \in \mathcal{X}_{\mathrm{adm}}} \mathrm{EI}(\boldsymbol{\xi}). \tag{14}$$

After deployment, the physical reward is added to $\mathcal{B}_n$ and the reward-transfer surrogate is updated. ~~The DT restricts and guides the search around π_{MB}, while the physical rewards allow RT to improve upon that policy.~~

Summary of the Reinforcement Twinning loop: ~~The RT closed-loop is summarized in~~ Fig. 1 ~~. The diagram represents one physical-digital iteration and makes explicit where information is created, where it is interpreted, and how it affects the next interaction with the physical environment~~shows one iteration of the RT loop, from the deployed policy to the choice of the next one. The selector e_n activates one of two learning objectives without changing the architecture of the loop. An information-oriented episode modifies the physical model through factual replay, whereas a performance-oriented episode leaves that model frozen and updates the reward-transfer surrogate. ~~In both cases, the physical environment is exposed only to a policy that has first been generated and screened in the DT.~~

The ~~actions carried out in each block can be summarized as follows:~~ blocks of Fig. 1 perform the following operations.

1) Policy deployment~~: the iteration starts from the parameters selected at the end of the previous DT evaluation. The first interaction~~ . The policy selected in the previous iteration is deployed; the first episode uses the baseline policy π_{BL}, ~~executed through the built-in PID controller, and provides~~ which sets the reference performance~~for comparison with the optimized policies. For $e_n = 1$, the deployed policy is an information-oriented policy $\pi_\iota(\cdot \mid \boldsymbol{\theta}_\iota^{(n)})$; for $e_n = 0$, it is a performance-oriented policy $\pi_\bullet(\cdot \mid \boldsymbol{\theta}_\pi^{(n)})$. This first block converts a design decision made in the DT into an executable command for the physical controller. The policy representation, its admissible parameter set, and all safety checks are application dependent, but the RT principle is unchanged: the DT proposes the interaction, while the physical environment supplies the evidence that determines whether the proposal was useful~~.

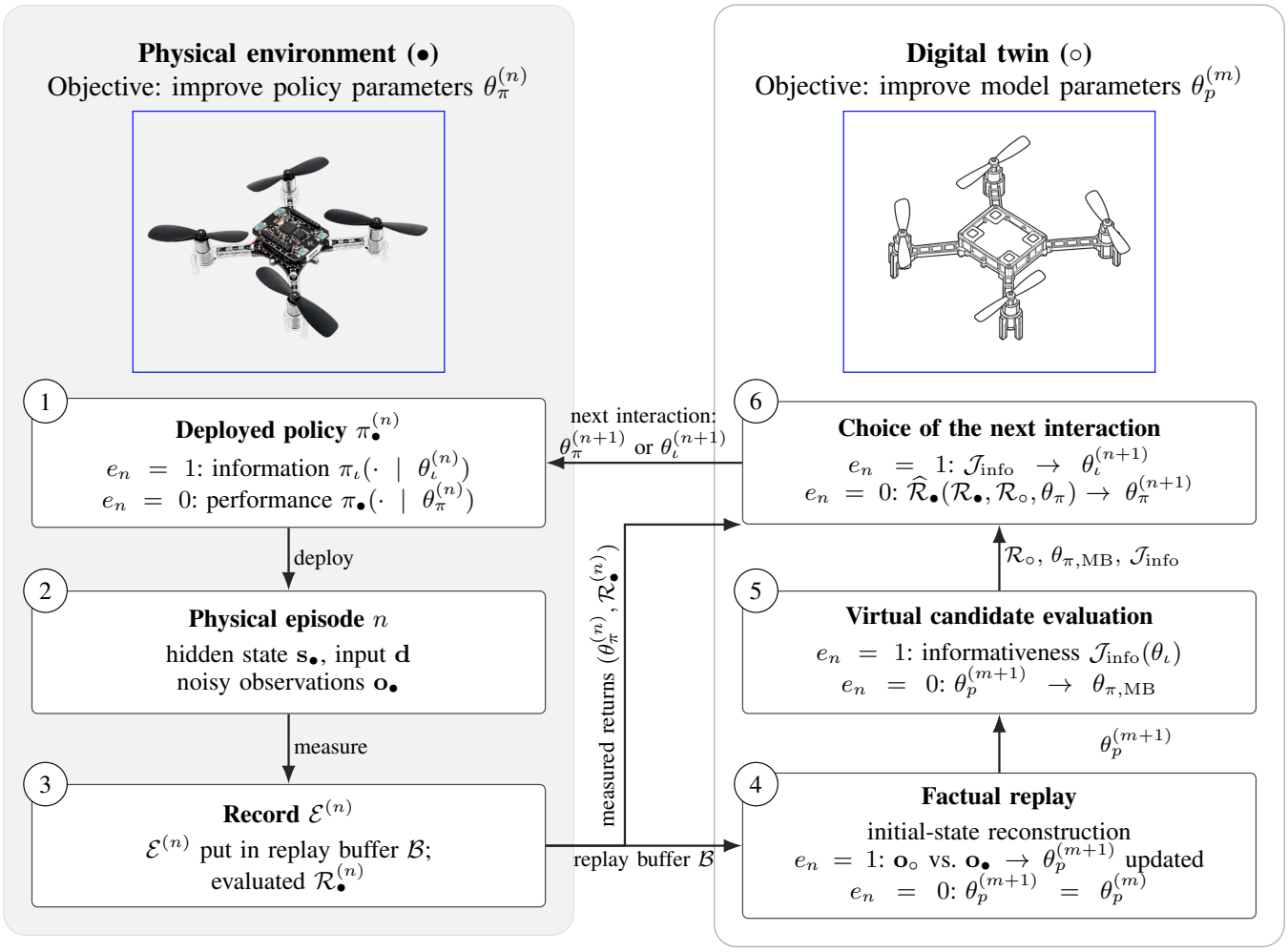


Fig. 1. ~~Reinforcement Twinning closed-loop~~ RT loop adopted in this work. The physical environment and its active ~~Digital Twin~~ DT evolve side by side through physical observations and virtual candidate evaluations~~, in order to achieve their respective objectives~~. The selector e_n distinguishes information-oriented ($e_n = 1$) and performance-oriented ($e_n = 0$) ~~interactions, keeping~~ episodes within a common loop~~for both cases~~. In the figure, $\mathcal{J}_{\text{info}}$ is the E-optimal criterion of Eq. (9) and $\widehat{\mathcal{R}}_\bullet$ the surrogate prediction of the physical reward, written $\mu_\bullet$ in Eq. (12).

2) Physical episode~~: the~~ . The physical environment evolves under its hidden dynamics, the deployed policy, and the exogenous input $\mathbf{d}$~~. The latter collects quantities that influence the episode without being selected by the policy, such as environmental disturbances, initial-condition variations, or changes in available actuation. The internal state $\mathbf{s}_\bullet$ is not generally known. RT receives~~ , and returns the noisy observations $\mathbf{o}_\bullet$ ~~, the commanded inputs, and the operating conditions that are needed to reconstruct the episode.This distinction is essential because the DT is updated from what the experiment can actually measure~~of Eq. (2).
3) Recording~~: the observations, deployed policy, operating condition, control signals, exogenous inputs, and evaluated physical reward are collected in the episode~~ . The record $\mathcal{E}^{(n)}$ ~~and~~ is appended to the replay buffer $\mathcal{B}$~~. The purpose of this passage is to preserve a common, traceable description of the interaction before the information and performance branches use it differently. The replay buffer therefore links all later virtual operations to the experiment that produced their data and permits a model update to be repeated on the complete cumulative~~ , so that every model update uses the complete dataset.
4) Factual replay~~: the DT reconstructs the initial state and~~ . For $e_n = 1$, the DT replays the recorded ~~policy under the measured operating conditions. When $e_n = 1$, DT observations $\mathbf{o}_\circ$ are compared with $\mathbf{o}_\bullet$~~ episode, and the mismatch ~~is differentiated through the closed-loop twin. The model parameters are then updated from~~ between $\mathbf{o}_\circ$ and $\mathbf{o}_\bullet$ updates $\boldsymbol{\theta}_p^{(m)}$ to $\boldsymbol{\theta}_p^{(m+1)}$ ~~using the cumulative replay buffer. When~~ . For $e_n = 0$, the model ~~parameters remain frozen, so that $\boldsymbol{\theta}_p^{(m+1)} = \boldsymbol{\theta}_p^{(m)}$. The new policy-reward pair bypasses identification and instead augments the dataset used to learn~~ stays frozen, and the new policy–reward pair enters the reward-transfer surrogate. ~~Factual replay thus has a different learning target in the two phases, while the recorded evidence entering the loop remains the same.~~
5) Virtual candidate evaluation~~: with the current model~~

parameters, the DT evaluates interactions that have not yet been carried out. In the information branch, information-oriented policiesare propagated through the closed-loop model, differentiated with respect to $\boldsymbol{\theta}_p$, and scored by the information they are expected to add to the weakest parameter direction. In the performance branch, the converged DT first provides the model-based policy π_{MB}. Candidate deviations around this policy are then simulated, rejected if they violate the DT safety checks , and assigned both . The DT simulates candidate policies. Information-oriented candidates are scored by their predicted information on $\boldsymbol{\theta}_p$. Performance-oriented candidates around π_{MB} that pass the DT checks receive a DT reward and a reward-transfer surrogate prediction of their physical reward. This block is the main computational advantage of RT: a large candidate set can be explored in the DT while only a small number of selected policies reaches the physical environmentMany candidates are evaluated virtually, but only the selected one is flown.

6) Choice of the next interaction: the final block converts the virtual evaluations into the next physical policy. For . Equation (9) ($e_n = 1$, Eq. (9) selects the candidate that maximizes the smallest eigenvalue of the cumulative Fisher matrix and therefore strengthens the least-observed joint parameter direction. For $e_n = 0$,) or Eq. (14) selects the candidate with the largest Expected Improvement, balancing a high predicted physical reward against uncertainty in the learned transfer. The selected parameters are returned to Block 1, the physical episode index advances from ($e_n = 0$) selects the next policy, n advances to $n + 1$, and the loop repeats. In the present work, the model-update index m advances only in the information phase; separating the two indices makes clear that collecting a new performance episode does not imply a new model update. returns to Block 1.

The selector value is scheduled within the test campaign. The loop first repeats Blocks 1-6 with $e_n = 1$ until the model satisfies the adopted convergence and uncertainty criteria. It then switches once to $e_n = 0$ and keeps the identified parameters fixed while the reward-transfer surrogate is refined. The following subsection specializes every element of this generic loop for the adopted test case.

B. Test Case Physical and Digital Environments

Experimental setup: The experiments are designed to reproduce a transition between out-of-ground-effect (OGE) and in-ground-effect (IGE) conditions over a finite obstacle while maintaining a constant forward flight speed. The experimental setup and adopted reference frame are illustrated in Fig. 2.

The campaign is performed using a Crazyflie 2.1+ MAV from Bitcraze flying in the x-z plane at $y = 0$ m. The horizontal reference frame retains its origin at axes keep the origin $x = y = 0$ of the positioning frame, whereas the vertical origin is placed at the upper surface of the obstacle, so that $z = 0$ at the obstacle heighta table whose top is 0.63 m above the floor. Positive z therefore measures the vertical distance of the vehicle from this surface , and is the height above this surface of the rotor plane, which lies 8.5 mm above the center of mass and 1.5 mm above the Lighthouse deck. With $R = 23$ mm the rotor radius, z/R is the corresponding nondimensional clearance, with $R = 23$ mm. Each test begins at $x = 0$ m, where the MAV reaches the prescribed altitude and hovers nondimensional clearance. After takeoff, the MAV moves to $x = 0.20$ m, climbs to the desired clearance z^*, and hovers there for 10 s. It then accelerates smoothly and reaches the desired forward speed around $x = 1.0$–1.2 m while maintaining the prescribed z/R. The region of interest used for comparison with the DT results spans $x = 1.70$–4.00 m. The MAV subsequently decelerates and lands around $x = 4.20$ m. Within the region of interest, the obstacle extends from $x = 2.20$ m to $x = 3.01$ m, corresponding to a streamwise length of 0.81 m, and it is 1.0 m wide. In the results, the streamwise coordinate is shifted to the beginning of the region of interest and normalized by R; consequently, this region covers $0 \leq x/R \leq 100$, while the obstacle occupies approximately $21.7 \leq x/R \leq 57.0$.

The Crazyflie 2.1+ is equipped with an onboard IMU sensor carries an onboard inertial measurement unit (IMU) for attitude control and a Lighthouse positioning deck for position control. Together with a With its 350 mAh battery, the drone mass is around its measured mass is $m = 32.42$ g. To provide a continuous position referenceFor continuous positioning, the Lighthouse deck is detected by two Lighthouse infrared receives the infrared sweeps of two Lighthouse base stations. Their placement is selected such that the MAV remains visible to at least one station throughout the complete trajectory. In the region directly above the obstacle, which constitutes the main area of interest for the GE experiments, the fields of view of the two stations intentionally overlap. This configuration reduces the probability of temporary tracking loss or a position-estimation disturbance caused by an interruption of the optical link between a Lighthouse station and the Lighthouse deck mounted on the Crazyflie. The Lighthousepositioning system provides sub-centimetre-level absolute accuracy and sub-millimetre repeatability. In flight experimentsoverlap reduces the risk of losing the position estimate when one optical link is interrupted. For Lighthouse V2, Taffanel et al. [55] reported an average positioning error of approximately 0.5 — 1 cm and an RMS position jitter between 0.5 and 1 mm , mean Euclidean errors of 2–4 cm in flight, with outliers up to about 5 cm, and a stationary jitter of 0.3–0.7 mm depending on the adopted estimator.

During each run, the Crazyflie firmware records the vehicle state, control response, and operating conditionsflight data are streamed by radio to the ground station (position every 20 ms, the other channels every 50 or 100 ms), where synchronized records are stored at about 10 Hz. The logged quantities include position and translational velocity, attitude and angular rates, acceleration, battery voltage, collective-thrust command, and the individual motor PWM pulse-width-modulation (PWM) commands. The complete record is retained for trajectory assessment and for reconstructing each physical

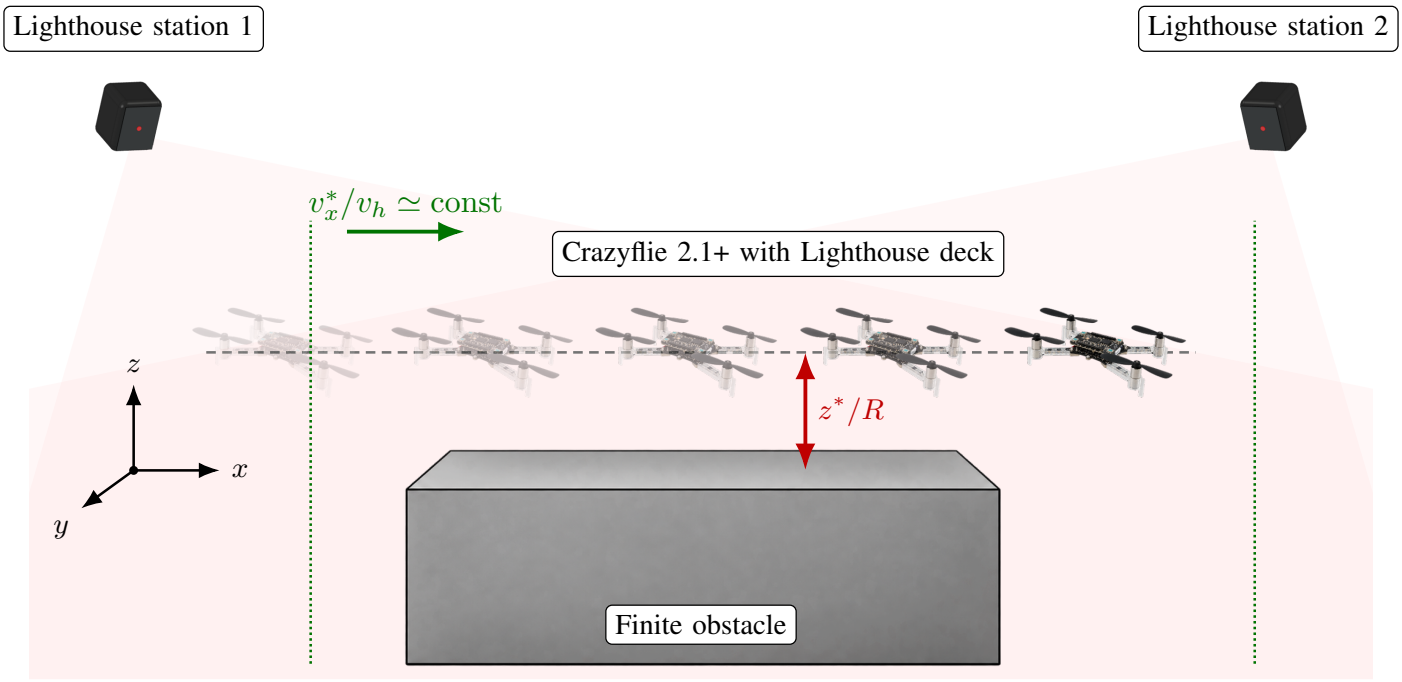


Fig. 2. Experimental setup, operating trajectory, Lighthouse coverage, and adopted reference frame.

episode in the DT. For GE identification, the physical and DT trajectories are aligned along the streamwise coordinate ~~, and the adjoint optimization minimizes the normalized vertical-position discrepancy over the obstacle region. The remaining channels support initialization and replay of the measured operating conditions, but~~ (Section II-C). The other logged channels serve trajectory assessment and do not enter the identification objective~~directly~~.

The Crazyflie is operated using its standard cascaded PID control architecture, as documented by Bitcraze [56]. During the crossing, the ground station sends a forward-velocity command scheduled on the measured x, a lateral velocity command that holds $y = 0$, and the altitude setpoint, every 20 ms. An outer position loop ~~tracks the desired position $\mathbf{x}^*$ and generates the corresponding velocity references $\mathbf{v}^*$~~converts the altitude setpoint into a vertical velocity reference. A velocity loop then regulates the translational motion, producing the desired roll and pitch angles together with the collective thrust command. These references are passed to the attitude controller, which determines the desired body angular rates. Finally, the inner angular-rate controller generates the roll, pitch, and yaw control contributions, which are combined with the collective thrust through the onboard motor mixer to determine the four individual motor commands. The ~~same control architecture and firmware logic are reproduced in the DT from the official implementation [57], ensuring consistency between the numerical and experimental closed-loop dynamics~~DT reproduces this command mode, the firmware logic [57], and the loop rates (100 Hz for position and velocity, 500 Hz for attitude and rates), so that the DT and the vehicle run the same controller. The flight script aborts a mission when the position log is older than 0.35 s or the end of the mission is not reached in time, and it rejects reference corrections larger than $4.55R$.

Digital Twin: The ~~Digital Twin developed in this work follows the modeling formulation proposed by Richardson et al. [52]for low-speed free flight~~DT follows the model of Richardson and Pretorius [52], validated by its authors in near-hover flight, and is tailored to ~~reproduce the dynamics of the Crazyflie~~the Crazyflie 2.1+ ~~employed in the experimental~~ of the campaign. The model is implemented as a differentiable physics-based simulator using the JAX library in Python and is used for both model identification and control optimization. The vehicle kinematics, rigid-body dynamics, propulsion system, ~~battery dependence,~~ actuator dynamics, and cascaded controller together form the nominal starting DT. ~~Their~~ The rigid-body, inertia, rotor-coefficient, actuator, and controller parameters are inherited ~~unchanged from Richardson et al. ; they are neither reidentified nor independently revalidated in the present campaign~~from Richardson and Pretorius. The vehicle mass was measured, and the effective command-to-thrust gain $T_{\max} = 0.1148$ N per motor was calibrated from the measured mass and the steady-hover collective command before the obstacle. No other parameter is reidentified. The finite-obstacle ground-effect correction is incorporated below as the unknown model component with associated $\boldsymbol{\theta}_p$ learned by RT.

The Crazyflie is modeled as a six-degree-of-freedom rigid body. Its state is described by position $\mathbf{x} = [x, y, z]^T$ and translational velocity $\mathbf{v} = [v_x, v_y, v_z]^T$ expressed in the inertial frame, the attitude quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]^T$, and the angular velocity $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$ expressed in the body-fixed frame. The rotational speed of rotor i is represented by Ω_i.

The translational dynamics evaluated at the center of mass

(CoM) are

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{v} \\ m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{F}_B \end{cases} \tag{15}$$

where $\mathbf{g} = [0, 0, -g]^T$, g is the gravitational acceleration, and $\mathbf{R} \in SO(3)$ transforms quantities from the body-fixed frame to the inertial frame. The total force in body coordinates is $\mathbf{F}_B = [0, 0, T]^T$, where T is the total thrust generated by the four rotors.

The rotational dynamics follow Euler's rigid-body equation,

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \boldsymbol{\tau} \tag{16}$$

where $\mathbf{I}$ is the inertia tensor and $\boldsymbol{\tau} = [\tau_x, \tau_y, \tau_z]^T$ is the total moment vector expressed in body coordinates. The vehicle attitude is propagated through the unit quaternion according to

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Lambda}(\boldsymbol{\omega})\mathbf{q} \tag{17}$$

where $\boldsymbol{\Lambda}(\boldsymbol{\omega})$ is the scalar-first quaternion kinematic matrix associated with the body angular velocity, using the convention adopted by Richardson ~~et al.~~and Pretorius [52]. The thrust and reaction torque generated by rotor i are modeled as

$$\begin{cases} T_i = k_T \Omega_i^2 \\ Q_i = k_Q \Omega_i^2 \end{cases} \tag{18}$$

where T_i and Q_i are the individual rotor thrust and torque and k_T and k_Q are the corresponding coefficients. The total thrust and moments follow from their signed combination. For the adopted quadcopter and motor convention, the roll (τ_R), pitch (τ_P), and yaw (τ_Y) moments are

$$\begin{cases} \tau_R = \frac{d}{2}\left(-T_1 - T_2 + T_3 + T_4\right) \\ \tau_P = \frac{d}{2}\left(-T_1 + T_2 + T_3 - T_4\right) \\ \tau_Y = \sum_{i=1}^{4} s_i Q_i, \end{cases} \tag{19}$$

where d is the ~~motor-to-motor distance~~ distance between adjacent motors and $s_i \in \{-1, +1\}$ identifies the rotation direction of rotor i.

~~The connection between the motor commands and propeller speed follows the Crazyflie propulsion characterization adopted by Richardson et al. The effect of battery state is included through the battery voltage $V_b = f(\mathrm{SOC})$, where SOC denotes the battery state of charge. A voltage correction factor is defined as $\eta_V = V_b / V_{\mathrm{ref}}$, where V_{ref} is the voltage of a fully charged battery. The rotor-speed command associated with the PWM input is corrected as $\Omega_{\mathrm{cmd}} = \eta_V \Omega_{\mathrm{PWM}}$~~With the firmware battery compensation enabled, each motor command $c_i \in [0, 65535]$ is a voltage-independent thrust request. The DT therefore maps it linearly to rotor thrust, $T_i = T_{\max} c_i / 65535$, and sets the commanded rotor speed $\Omega_{i,\mathrm{cmd}} = (T_i/k_T)^{1/2}$. The realized ~~rotor~~ speed follows this command through a first-order ~~actuator model~~ lag with time constant $\tau_{\mathrm{rpm}} = 0.020$ s ~~. This value is inherited from the Richardson model and is not identified in the present work~~and a rate limit of 2×10^5 rpm/s.

~~The main physical parameters adopted in the DT are taken from the formulation of [52] and are summarized in~~ Table I ~~. The nominal propulsion model retains the battery-voltage correction $V_b = f(\mathrm{SOC})$. However, both physical and DT missions start with a fully charged battery and last only a few seconds; therefore, the within-mission change of state of charge is neglected. The vehicle dynamics are coupled with the same cascaded PID architecture used in the physical environment, providing the nominal closed-loop model on which the ground-effect correction is identified~~lists the DT parameters; the DT also uses the products of inertia of Richardson and Pretorius, up to 1.80×10^{-6} kg m^2. The dynamics are integrated with a fourth-order Runge–Kutta scheme at a 2 ms step, and aerodynamic drag is neglected.

TABLE I

~~BASELINE~~ DT PARAMETERS~~ADOPTED~~. THE MASS IS MEASURED, R IS THE PROPELLER RADIUS, AND $T_{\max}$ IS CALIBRATED FROM HOVER; THE OTHER VALUES ARE THOSE OF RICHARDSON ~~ET AL.~~AND PRETORIUS [52].

Parameter	Value
m	0.03242 kg
R	0.023 m
d	0.065 m
I_{xx}	16.57×10^{-6} kg m^2
I_{yy}	16.56×10^{-6} kg m^2
I_{zz}	29.26×10^{-6} kg m^2
k_T	2.88×10^{-8} N s^2/rad^2
k_Q	7.24×10^{-10} N m s^2/rad^2
τ_{rpm}	0.020 s
~~V_{ref}~~ $T_{\max}$	~~4.20 V~~ 0.1148 N

Ground-effect correction: The ~~described nominal DT reproduces the vehicle dynamics in free flight and was validated in those conditions by Richardson et al. [52]. The nominal DT, however,~~ nominal DT does not account for the ~~additional rotor-surface~~ rotor–surface interaction generated as the vehicle ~~closely~~ crosses the finite obstacle at low clearance. This interaction is ~~introduced through a deliberately simple physical model~~ modeled with a low-order law inspired by the ~~analytical formulation of Cheeseman and Bennett~~ Cheeseman–Bennett relation for an isolated rotor~~in hovering~~ [7]. For a rotor i of radius R at a clearance z_i from the obstacle, the in-ground-effect thrust $T_{i,\mathrm{IGE}}$ is obtained from the nominal out-of-ground-effect thrust $T_{i,\mathrm{OGE}}$ through the amplification factor f_{GE},

$$T_{i,\mathrm{IGE}} = T_{i,\mathrm{OGE}}\, f_{\mathrm{GE}}(z_i; \boldsymbol{\theta}_p). \tag{20}$$

The amplification factor is modeled as

$$f_{\mathrm{GE}}(z_i; \boldsymbol{\theta}_p) = 1 + A\left(\frac{R}{z_i}\right)^p, \qquad \boldsymbol{\theta}_p = [A,\, p]^\top. \tag{21}$$

Here, A controls the magnitude of the thrust augmentation, while p determines how rapidly the interaction decays with increasing clearance. ~~The original~~ For a hovering rotor at constant power, Cheeseman and Bennett ~~formulation rearranged with the same unknowns would have retained~~ obtained $T_{\mathrm{IGE}}/T_{\mathrm{OGE}} = [1 - (R/4z)^2]^{-1}$ [7], whose first-order expansion gives $A = 1/16$ and $p = 2$. These values initialize the identification~~in the present work, while both coefficients remain free to adapt to the present vehicle and operating conditions~~. The DT applies f_{GE} to the thrust at the actual rotor speed, not at constant power, so the identified A and p are effective coefficients of the vehicle in closed loop.

~~This model is deliberately low order: ground effect is treated as a static~~ The model is static. It treats ground effect as a function of the instantaneous rotor clearance ~~. Measurements by~~ only. Kan et al. [17] ~~indicate thatvelocity dependence is limited over the low-speed range considered here, but it is not assumed to vanish. The formulation~~ found that, at low and medium speeds, ground effect depends much more on clearance than on forward speed. Up to the 0.5 m/s flown here, the forward-flight term of the Cheeseman–Bennett relation also changes the augmentation by less than 2%. The model therefore neglects the speed dependence. It also omits the transient development and decay of the ~~rotor-surface interaction as the vehicle crosses~~ rotor–surface interaction at the obstacle edges [8], and it is monotone in clearance, whereas a negative ground effect near $z/R \approx 1.5$ has been reported for a closely spaced quadrotor [14]. The identified law is therefore an effective model over the tested clearances, not a general quadrotor law. These omissions preserve a compact model that can be identified and differentiated efficiently, while leaving a structural discrepancy that is addressed later through the reward-transfer surrogate.

Equation (21) describes the clearance dependence above an extended surface. The obstacle considered here occupies only a finite streamwise interval, so the augmentation must also depend on the position of each rotor relative to its edges. ~~It~~ The augmentation is therefore weighted by the fraction $\chi_i(x) \in [0,1]$ ~~, defined as the fraction~~ of rotor disk i that overlaps the obstacle, in line with measurements on a partially covered rotor whose thrust correlates with the overlapping disk area [15]. The corrected rotor thrust becomes

$$T_{i,\mathrm{corr}} = T_{i,\mathrm{OGE}}\left[1 + \chi_i(x)\left(f_{\mathrm{GE}}(z_i;\boldsymbol{\theta}_p) - 1\right)\right], \tag{22}$$

where ~~$T_{i,\mathrm{OGE}}$ is the nominal rotor thrust outside ground effect. The~~ the overlap fraction is zero when the rotor lies outside the obstacle footprint and reaches unity when its disk is fully above it. Its continuous variation across the leading and trailing edges avoids discontinuous force changes and ~~naturally~~ accounts for the four rotors entering and leaving the obstacle ~~region~~ at different streamwise positions. It weights each rotor independently, whereas free-flight tests near a building-like edge found that full ground effect develops only after most of the vehicle has crossed the edge [16].

C. Test Campaign Overview

Following Kan et al. [17], the desired forward ~~velocity~~ speed v_x^* is normalized by the ideal induced velocity in hover,

$$v_h = \sqrt{\frac{T_{h,r}}{2\rho_a \pi R^2}} = \sqrt{\frac{mg}{8\rho_a \pi R^2}}, \tag{23}$$

where $T_{h,r} = mg/4$ is the hover thrust ~~required from each rotor ,~~ of each rotor and ρ_a is the air density~~, and~~ . With m and R ~~is the rotor radius. With $m = 32.42$ g, $R = 0.023$ m,~~ from Table I and $\rho_a = 1.225$ kg/m^3, Eq. (23) gives $v_h \simeq 4.42$ m/s. An operating condition is henceforth denoted by the ordered pair $(z^*/R,\ v_x^*/v_h)$.

The ~~test campaign follows the sequential RT schedule introduced above. All tests use the same physical environment,~~ ~~DT structure, reference frame, and data-processing procedure described in Sec. II-B. The~~ nominal operating condition is $(z^*/R,\ v_x^*/v_h) = (1.5, 0.079)$. Unless stated otherwise, policy design, model identification, and reward evaluation refer to this condition. Table II lists the policies flown and the number of flights. The campaign is organized into five consecutive blocks, introduced below together with the notation used in the results.

TABLE II
POLICIES AND FLIGHTS OF THE CAMPAIGN. THE RECORD OF EACH POLICY IS THE MEAN OF ITS RETAINED FLIGHTS.

Stage and policies	Policies	Flights each	Flights
Baseline π_{BL}	1	3	3
Information π_{INF}	$10 + 2^a$	3–4	37
Model-based π_{MB}	1	3	3
Interpolation $\pi_{\lambda,i}$	2	4	8
Sobol $\pi_{\mathrm{a},i}$	7	3–5	27
EI $\pi_{\mathrm{EI},i}$	10	3–6	37
At $(2.5, 0.045)$	3	2–3	7
At $(1, 0.113)$	5	2–4	15
Total	41		137

[a] Archived records of an earlier campaign.

Baseline policy: The ~~campaign starts from the built-in cascaded PID controller~~ baseline policy π_{BL} is a constant reference at the nominal condition ~~$(z^*/R,\ v_x^*/v_h) = (1.5, 0.079)$. This policy is denoted by π_{BL} and is repeated experimentally~~ , tracked by the onboard cascaded PID controller. It is flown repeatedly to quantify run-to-run variability. Its mean response and RMS band provide the reference trajectory statistics, the first record in the replay buffer, and the baseline ~~tracking cost used to normalize the rewards reported later.Since π_{BL} reacts only after the vehicle departs from the desired clearance, it also exposes the ground-effect excursion that the subsequent predictive policies are designed to anticipate~~cost J_{BL} used in the normalized score of Eq. (45).

Information phase: The objective of the information phase is to design and perform trajectories that improve the identification of the unknown parameters $\boldsymbol{\theta}_p = [A, p]$ of the ground-effect model described by Eqs. (20) and (21). Each ~~trajectory deployed in the physical environment is replayed through the differentiable~~ episode is replayed in the DT with the ~~same referenceand reconstructed initial condition used experimentally~~deployed reference, starting at $x = 1.70$ m from the steady out-of-ground-effect state at the commanded clearance and speed of the record. Factual replay identifies ~~the coefficient pair~~ A and p from the accumulated tracking response~~. The subsequent DT design searches for a safe altitude reference that adds information along the least-observed parameter direction, thereby defining the next experiment. The two operations are repeated sequentially, with each new flight updating both the identified parameters and the information available for the next design.~~

~~The model is initialized from the Cheeseman-Bennett coefficients, $A = 1/16$ ~~, and ~~$p = 2$~~the DT then designs the next experiment.

Starting from the baseline record, twelve information-

oriented ~~policies~~ records $\pi_{\mathrm{INF}}^{(n)}$ are added ~~sequentially~~ to the cumulative dataset. ~~The first six policies are designed over the broad admissible clearance range $0.5 < z/R < 6$. This first stage deliberately~~ Ten were designed in this campaign in four batches, the last of which kept a single policy. Six span the broad range $0.5 \le z/R \le 6$ of the reference, and four the low-clearance range $0.3 \le z/R \le 1.5$. The other two are low-clearance records of an earlier campaign at $z^*/R = 1$ and $v_x^*/v_h = 0.091$ and 0.102, of which only the mean trajectories are available. The broad stage explores a wide set of ~~altitude histories: it~~ clearance histories. It establishes the global scale and decay of the interaction without assuming ~~in advance~~ that the most useful data ~~must~~ come from the smallest admissible clearance, and it ~~verifies the response of~~ tests the evolving DT over the ~~complete safe operating~~ whole admissible range.

~~By the end of the sixth information-oriented episode, new policies in the broad domain~~ In the post-campaign evaluation of Section III-B, the broad-domain episodes after the first still increase the cumulative Fisher information, but ~~their marginal E-optimal contribution has become modest compared with the first information-oriented interaction. This behavior is physically consistent~~ each adds much less than the first did. This agrees with Eq. (21)~~: at~~. At larger clearances, f_{GE} approaches unity and its sensitivity to both A and p decreases. ~~The admissible design domain is therefore restricted to $0.5 < z/R < 1.5$ for the remaining six policies, defining the near-wall domain. The restriction is an adaptive design decision, not a rejection of the broad-domain data.~~ After the sixth episode, the design domain of the reference is restricted to $0.3 \le z/R \le 1.5$, the low-clearance domain. All preceding episodes remain in the replay buffer, while the new candidates concentrate the available experimental effort where the identified law is steepest and the weak parameter direction can be excited more effectively. ~~After every flight, the new record is appended to the cumulative dataset and the identification~~ During the campaign, the model was updated after each batch. In the post-campaign identification of Section III-B, the records are appended one at a time, and each fit is warm-started from the preceding ~~converged coefficients~~estimate.

The information phase begins with factual replay and adjoint identification. For physical episode n, the ~~experimental~~ record provides the ~~measured altitude-tracking error, the deployed reference policy, and the reconstructed initial condition~~deployed reference and the measured clearance error $e_{z,n}^{\bullet}(x) = z(x) - z^*$. The DT replays the same episode with the current model parameters $\boldsymbol{\theta}_p$ ~~. The experimental and virtual error histories~~ and gives the corresponding error $e_{z,n}^{\circ}(x;\boldsymbol{\theta}_p)$. Both are evaluated at the ~~same~~ K_n ~~spatial positions encountered during the flight~~ samples of the measured record above the obstacle, $2.20 \le x \le 3.01$ m, and collected in the vectors

$$\mathbf{e}_{z,n}^{\bullet} = \left[e_{z,n}^{\bullet}(x_{n,1}), \ldots, e_{z,n}^{\bullet}(x_{n,K_n})\right]^{\mathsf{T}}, \tag{24}$$

$$\mathbf{e}_{z,n}^{\circ}(\boldsymbol{\theta}_p) = \left[e_{z,n}^{\circ}(x_{n,1};\boldsymbol{\theta}_p), \ldots, e_{z,n}^{\circ}(x_{n,K_n};\boldsymbol{\theta}_p)\right]^{\mathsf{T}}. \tag{25}$$

With N episodes in the replay buffer, the identification loss is

$$J_p(\boldsymbol{\theta}_p) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{K_n}\left\|\frac{\mathbf{e}_{z,n}^{\circ}(\boldsymbol{\theta}_p) - \mathbf{e}_{z,n}^{\bullet}}{R}\right\|_2^2. \tag{26}$$

The factor $1/K_n$ gives every episode equal weight irrespective of its number of samples or duration.

The loss is minimized by differentiating the complete time-discrete closed-loop rollout in reverse mode. This is the discrete-adjoint formulation of the identification problem: the loss sensitivity is propagated backward through the vehicle dynamics, actuator model, controller states, and ground-effect correction. At optimization step t, the resulting gradient is

$$\mathbf{g}_t = \nabla_{\boldsymbol{\theta}_p} J_p(\boldsymbol{\theta}_{p,t}) = \begin{bmatrix}\dfrac{\partial J_p}{\partial A} & \dfrac{\partial J_p}{\partial p}\end{bmatrix}^{\mathsf{T}}. \tag{27}$$

It accounts for the direct influence of A and p on the aerodynamic forces and for the trajectory changes produced through the closed feedback loop. Both derivatives are obtained in one backward pass, without a separate perturbed simulation for each parameter.

Adam [58] uses this gradient to update the parameters over successive optimization steps. It retains a first moment $\mathbf{m}_t$, which smooths the gradient direction, and a second moment $\mathbf{v}_t$, which tracks its component-wise squared magnitude:

$$\mathbf{m}_t = \beta_1\mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t, \tag{28}$$

$$\mathbf{v}_t = \beta_2\mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^{\odot 2}. \tag{29}$$

The two moments allow the step size to adapt separately to A and p, whose gradients can have different scales. After the standard bias correction, the next parameter vector is

$$\boldsymbol{\theta}_{p,t+1} = \Pi_{\Theta}\left[\boldsymbol{\theta}_{p,t} - \eta\frac{\widehat{\mathbf{m}}_t}{\sqrt{\widehat{\mathbf{v}}_t} + \epsilon}\right], \tag{30}$$

where η is the ~~chosen~~ learning rate, ϵ prevents division by zero, and Π_{Θ} ~~keeps the solution inside the physically admissible set Θ~~projects onto the admissible set $\Theta = \{0 \le p \le 6,\ A \ge -0.95\,(0.5)^p\}$, which keeps $f_{\mathrm{GE}} \ge 0.05$ for $z_i \ge 0.5R$. In the present campaign, $\eta = 5 \times 10^{-3}$ and convergence is assessed over windows of 25 optimization steps. The relative-loss, coefficient-range, and projected-gradient tolerances are 2×10^{-4}, 10^{-3}, and 5×10^{-3}, respectively. The optimization is considered converged only when all three conditions remain satisfied over a complete monitoring window, after at least 100 steps. When a new physical episode is added, the next fit is warm-started from the previously converged parameters, while the Adam moments are reset because the cumulative objective has changed. A cold start from $(A, p) = (0, 0)$ on the complete dataset converges to $(0.120, 1.970)$, against $(0.121, 1.980)$ with warm starts.

The converged estimate $\widehat{\boldsymbol{\theta}}_p$ is then used to determine which experiment should be performed next. Local parameter uncertainty is approximated from the curvature of the identification loss [59]. Let

$$\mathbf{H} = \nabla^2 J_p(\widehat{\boldsymbol{\theta}}_p) = \mathbf{Q}\,\mathrm{diag}(\lambda_1, \lambda_2)\mathbf{Q}^{\mathsf{T}}. \tag{31}$$

To avoid a singular inverse along a locally flat direction, the eigenvalues are floored at $\lambda_H = 10^{-6}$. The resulting local inverse-Hessian approximation is

$$\begin{aligned}\boldsymbol{\Sigma}_{\theta} \simeq{}& \max\left[J_p(\widehat{\boldsymbol{\theta}}_p), 10^{-12}\right]\mathbf{Q} \\ &\times \mathrm{diag}\left[\max(\lambda_1, \lambda_H)^{-1}, \max(\lambda_2, \lambda_H)^{-1}\right]\mathbf{Q}^{\mathsf{T}}.\end{aligned} \tag{32}$$

This matrix is used as a local curvature-based uncertainty approximation, not as a calibrated finite-sample confidence region. A flat direction of the loss produces a larger local uncertainty. For a model output $g(\boldsymbol{\theta}_p)$, such as f_{GE} at a prescribed clearance, the corresponding linearized uncertainty indicator is

$$\sigma_g^2 = \nabla_{\boldsymbol{\theta}_p} g^{\mathsf{T}} \boldsymbol{\Sigma}_\theta \nabla_{\boldsymbol{\theta}_p} g. \tag{33}$$

~~Thus, A and p may retain a strongly elongated and correlated local uncertainty region even when the resulting ground-effect curve is tightly constrained.~~ The quantity $2\sigma_{f_{\mathrm{GE}}}/f_{\mathrm{GE}}$ reported later is the propagated relative uncertainty of the total thrust factor. It is not interpreted as an exact 95% confidence interval, and its decrease at large clearance partly reflects that $f_{\mathrm{GE}} \to 1$.

Candidate altitude-reference policies are then generated inside the admissible design space and simulated with the converged DT. Because A and p have different numerical scales, the sensitivity columns are normalized with the local parameter scales

$$\mathbf{D}_\theta = \mathrm{diag}\left(\sqrt{[\boldsymbol{\Sigma}_\theta]_{AA}}, \sqrt{[\boldsymbol{\Sigma}_\theta]_{pp}}\right), \qquad \overline{\mathbf{S}}_n = \frac{\partial \mathbf{e}_{z,n}^{\circ}}{\partial \boldsymbol{\theta}_p} \mathbf{D}_\theta. \tag{34}$$

The eigenvalues can therefore be interpreted for perturbations of one local parameter standard deviation, rather than being dominated by the raw numerical scale of either coefficient. Under an independent, homoscedastic residual approximation, a common normalized residual variance is estimated as $\sigma_e^2 = \max[J_p(\widehat{\boldsymbol{\theta}}_p), 10^{-12}]$. The Fisher-information proxy accumulated from the N completed experiments is

$$\mathbf{F}_{1:N} = \sum_{n=1}^{N} \frac{\overline{\mathbf{S}}_n^{\mathsf{T}} \overline{\mathbf{S}}_n}{K_n \sigma_e^2}, \tag{35}$$

where K_n prevents a longer episode from appearing more informative solely because it contains more samples. Large eigenvalues of $\mathbf{F}_{1:N}$ correspond to locally scaled parameter combinations that the accumulated experiments constrain strongly; its smallest eigenvalue describes the least-observed joint parameter direction. Since spatial residuals can remain correlated, this matrix is ~~used as a comparative design~~ a comparative metric rather than ~~as~~ an exact statistical Fisher matrix. ~~During sequential design it is evaluated with the current converged DT. For the post-campaign comparison reported later,~~ It is evaluated after the campaign, for every episode and for the repeated-baseline counterfactual~~are reevaluated using~~, with the final DT, the final parameter scaling, and the same residual variance, so ~~their~~ that the cumulative values are directly comparable.

Each virtual candidate policy π produces a predicted Fisher contribution $\mathbf{F}(\pi)$. The next information-oriented policy is selected with the E-optimal criterion

$$\pi_{\mathrm{INF}}^{(N+1)} = \arg\max_{\pi \in \Pi_{\mathrm{adm}}} \lambda_{\min}(\mathbf{F}_{1:N} + \mathbf{F}(\pi)), \tag{36}$$

which ~~selects the safe candidate predicted to increase the Fisher information along its weakest direction. The criterionalready favors sensitivities that complement the accumulated dataset. A separate novelty check may nevertheless select a slightly less informative candidate when its predicted response is sufficiently different from the experiments already performed. Candidates that violate state, reference-rate, acceleration, actuator, or clearance constraints are discarded before deployment.~~

~~After the selected flight, its record is added to the replay buffer and the adjoint-Adam identification is repeated. The inner Adam loop is stopped by the numerical convergence conditions defined above. The complete information phase is terminated only when adding new episodes no longer changes the converged values of A and p appreciably and no longer produces a meaningful reduction in the magnitude of their estimated correlation~~is the idealized criterion. The campaign used a batch approximation of it. Each design iteration produced three policies from 68 candidates, 64 scrambled Sobol points and four structured shapes. The candidates were screened in the DT for reference rate ($|v_z| \le 0.5$ m/s), reference acceleration ($|a_z| \le 2$ m/s^2), completion of the rollout, and a lowest-rotor clearance of at least $0.3R$. The first policy maximizes $\lambda_{\min}$ of its own predicted Fisher matrix among the most novel half of the 30% most informative candidates, where novelty is the distance of the predicted response from the measured trajectories. The second maximizes $\lambda_{\min}$ of the joint matrix with the first, and the third maximizes the novelty of the predicted response after the trailing edge, subject to minimum information levels. The predicted Fisher matrices are averaged over up to five sigma-point samples of $\boldsymbol{\theta}_p$. The accumulated information $\mathbf{F}_{1:N}$ of past experiments is not used in the design, where past trajectories enter through the novelty filter.

The identification is then repeated on the enlarged replay buffer, and the information phase ends with the criterion of Section II-A.

Model-based policy: After identification, A and p are frozen~~and the specialized performance loop uses the converged DT~~, and the DT is used to improve altitude tracking across the ~~finite~~ obstacle. The ~~control policy is implemented as~~ policy is a spatial reference ~~governor above the unchanged cascaded PID controller, so the computationally expensive prediction and optimization remain offlinewhile the deployed policy is a lightweight lookup table. First the DT produces the model-based policy π_{MB}; physical rewards around that policy then train the reward-transfer surrogate used by Expected Improvement to design improved policies. The policy is the spatial altitude reference supplied to the governor~~correction, computed offline. During the flight, the ground station interpolates it at the measured streamwise position and sends the corrected altitude setpoint z_π to the unchanged cascaded PID controller,

$$z_\pi(x) = z_{\mathrm{BL}}(x) + \Delta z_\pi(x), \tag{37}$$

where z_{BL} is the nominal mission reference altitude which is constant along x. The ~~governor~~ correction modifies only the vertical ~~set point~~ setpoint, as a function of ~~measured longitudinal~~ the measured streamwise position. The horizontal trajectory, attitude stabilization, motor mixing, and onboard PID gains remain unchanged. ~~This spatial representation is preferred over the time representation as it prevents small DT-experiment timing differences from desynchronizing~~ A

spatial rather than temporal parameterization prevents small timing differences between DT and experiment from shifting the anticipatory command ~~from~~ relative to the obstacle.

At the nominal operating condition, the reference correction is parameterized at ~~control points separated by one rotor radius, $\Delta x = R$, over the complete crossing and interpolated between consecutive points~~$N_x = 81$ nodes spaced by 25 mm ($1.09R$) over $2.00 \le x \le 4.00$ m, linearly interpolated between them, and zero outside this interval. Its amplitude is bounded by $|\Delta z| \le 3R$, while the predicted lowest rotor must retain a clearance of at least $0.5R$ from the local surface.

The model-based policy ~~is first obtained through model-predictive reference synthesis. The model-based policy~~ π_{MB} is obtained offline from a finite-horizon nonlinear prediction of the complete closed-loop mission. At every optimizer step, the DT propagates the vehicle, propulsion, aerodynamic model, PID states, and reference ~~governor~~ correction over the entire prediction interval. Reverse-mode differentiation supplies the gradient with respect to all spatial reference values, and Adam updates their amplitudes subject to bound and endpoint projections. ~~The optimized policy is therefore computed before the flightand then executed online as a lightweight interpolation table. In this work, *model predictive control* refers to this model-predictive reference synthesis; a receding-horizon optimizer is not run onboard~~No optimization runs during the flight.

Performance is ~~evaluated against the original mission reference used by the nominal cascaded PID controller~~measured against the nominal reference $z_{\mathrm{BL}} \equiv z^*$. The clearance z_k is taken from the repetition-averaged trajectory at the $N_{\mathcal{R}}$ samples x_k in the window $\Omega_{\mathcal{R}} = [2.20, 3.70]$ m ($21.7 \le x/R \le 87.0$), with $N_{\mathcal{R}} \simeq 42$ at the nominal condition. DT rewards are evaluated at the same samples by factual replay with the final coefficients. The tracking cost and ~~corresponding reward are~~ reward are

$$J_{\mathrm{trk}}(\pi) = \frac{1}{N_{\mathcal{R}}} \sum_{x_k \in \Omega_{\mathcal{R}}} \left[\frac{z_k - z_{\mathrm{BL}}(x_k)}{R} \right]^2, \qquad \mathcal{R}(\pi) = 1 - J_{\mathrm{trk}}(\pi). \tag{38}$$

The reported policy is optimized with the single final identified DT; no model-uncertainty ensemble is used. The objective augments the tracking term with amplitude, curvature, and clearance penalties,

$$J_{\mathrm{MB}} = J_{\mathrm{trk}} + \frac{0.10}{N_x} \left\| \frac{\Delta z_\pi}{R} \right\|_2^2 + \frac{2.0}{N_x - 2} \left\| \mathbf{D}^2 \frac{\Delta z_\pi}{R} \right\|_2^2 + 10^5 \Phi_{\mathrm{clr}}(\pi), \tag{39}$$

where J_{trk} is evaluated over $2.20 \le x \le 4.00$ m, $\mathbf{D}^2$ is the ~~spatial~~ second-difference operator on the nodes, and Φ_{clr} is the mean squared violation of the $0.5R$ minimum clearance of the lowest rotor while ~~it~~ the vehicle is above the obstacle. The amplitude term limits control authority, the curvature term suppresses spatially sharp commands, and the clearance term ~~enforces safety on the tilted rotor geometry~~ applies the $0.5R$ limit to each tilted rotor rather than only ~~on the vehicle~~ to the center of mass. Adam runs for 150 iterations with learning rate 10^{-3}, starting from $\Delta z_\pi = 0$. The final policy is replayed once more in the DT and rejected if the hard clearance check fails.

Performance phase: The performance search is conducted at the nominal operating condition with the identified parameters frozen. Two interpolation policies, denoted collectively by $\pi_{\lambda,i}$ with $\lambda \in \{1/3, 2/3\}$, are first tested between π_{BL} and π_{MB} to establish the global relation between the DT and physical rewards. The search is then expanded around π_{MB} using three smooth spatial basis functions. A scrambled Sobol library~~of 2000~~ [60] of 2048 coefficient vectors is screened in the DT, and seven admissible ~~, mutually distinct~~ policies $\pi_{\mathbf{a},i}$ ~~are deployed~~ , chosen to span different amplitude combinations, are flown to train the ~~local~~ GP discrepancy. ~~Subsequent~~ Ten policies $\pi_{\mathrm{EI},i}$ are ~~selected sequentially by Expected Improvement; after each flight, the~~ then selected by EI and flown, and each physical reward is ~~appended to the reward-transfer~~ added to the surrogate dataset. The policy with the largest nominal-condition physical reward is denoted by $\pi^\star_{\mathrm{EI}}$.

~~Physical rewards are then used to refine this model-based policy and compensate for the remaining model discrepancy. Reinforcement Twinning uses these measurements to correct the local policy ranking predicted by the DT. A new candidate~~ A candidate policy is written as

$$\Delta z_\pi(x) = \Delta z_{\mathrm{MB}}(x) + R\, \mathbf{b}^{\mathsf{T}}(x)\mathbf{a}, \tag{40}$$

where ~~the smooth basis functions~~$\mathbf{a} = [a_A, a_B, a_C]^\top \in [-1, 1]^3$ and $\mathbf{b}(x)$ ~~span the control-relevant portions of the mission and $\mathbf{a} = [a_A, a_B, a_C]^\top$ is the bounded vector of policy-modification coefficients. A space-filling library is first evaluated in the DT; the same~~ holds three basis functions. Each is a Gaussian centered at $x = 2.50$, 2.95, and 3.30 m ($x/R \simeq 35$, 54, and 70 in the shifted coordinate of the figures), with standard deviation 0.25, 0.20, and 0.25 m. It is multiplied by $\sin^2[\pi(x - x_0)/L]$, with $x_0 = L = 2.00$ m, and normalized to unit peak, so that a_j is a peak amplitude in rotor radii. The reference and rotor-clearance checks used for ~~the MB solution~~ π_{MB} define the admissible set ~~$\mathcal{A}_{\mathrm{safe}}$~~$\mathcal{A}_{\mathrm{adm}}$.

The affine component of the reward-transfer surrogate is fitted by ordinary least squares to the paired DT and physical rewards. The residual is modeled by a zero-mean Matérn-3/2 GP. At the nominal operating condition, its input is ~~the three-component vector~~ $\mathbf{a}$, with ~~each coordinate scaled by $0.30R$~~a length scale of 0.30 in each coordinate. The signal variance is ~~initialized from~~ set to the empirical variance of the affine residuals~~and is bounded below by~~ , with a lower bound of 0.05^2, and no hyperparameter is optimized. The rewards are computed from the ~~cleaned mean~~ repetition-averaged trajectories and are treated as deterministic ~~campaign~~ observations; a diagonal jitter of 10^{-8} is added only for numerical stability. Accordingly, the GP posterior standard deviation quantifies interpolation uncertainty within this surrogate and not run-to-run experimental dispersion. The resulting relation is

$$\mathcal{R}_\bullet = \gamma_0 + \gamma_1 \mathcal{R}_\circ + \delta(\mathbf{a}, \boldsymbol{\chi}) + \varepsilon, \tag{41}$$

where δ is the GP discrepancy and $\boldsymbol{\chi}$ contains optional operating conditions expressed by $(z^*/R,\, v_x^*/v_h)$. ~~Two policies with similar DT performance can receive different physical corrections. Its~~ The posterior mean and variance of δ are updated after every performance-oriented episode. Expected Improvement is used with zero exploration margin, ~~$\xi = 0$~~$\zeta = 0$, and selects the next experiment,

$$Z(\mathbf{a}) = \frac{\mu_\bullet(\mathbf{a}) - \mathcal{R}_\bullet^+}{\sigma_\bullet(\mathbf{a})}, \tag{42}$$

$$\mathrm{EI}(\mathbf{a}) = \left(\mu_\bullet - \mathcal{R}_\bullet^+\right)\Phi(Z) + \sigma_\bullet \phi(Z), \tag{43}$$

$$\mathbf{a}_{n+1} = \arg\max_{\mathbf{a}\in\mathcal{A}_{\mathrm{adm}}} \mathrm{EI}(\mathbf{a}), \tag{44}$$

with ~~zero improvement assigned to a vanishing posterior variance . The first term exploits candidates with a high predicted physical reward, whereas the second rewards unresolved but potentially useful regions. The model-based policy remains the reference throughout the search; data are used to learn the GP discrepancy that affects physical performance~~the posterior variance floored at 10^{-8}. EI is maximized over the Sobol library, six points at ± 0.25 on the axes, the coefficients of the policies already flown, and 128 Gaussian perturbations ($\sigma = 0.10$) of the four best, all screened in the DT.

For transfer across operating conditions, the DT and physical performance are normalized by the corresponding condition-specific baseline cost,

$$\mathcal{S} = 1 - \frac{J_\pi}{J_{\mathrm{BL}}} = 1 - \frac{1-\mathcal{R}_\pi}{1-\mathcal{R}_{\mathrm{BL}}}, \tag{45}$$

and $\boldsymbol{\chi}$ is augmented with the operating condition. ~~Policy coordinates are defined relative to the local MB deformation. Consequently, a~~ To transfer a policy, each coefficient is scaled by the ratio s_j of the root-mean-square amplitude of the condition-specific model-based correction to that of the nominal one, both weighted by b_j, $a_j^{\mathrm{tr}} = s_j a_j^\star$, where $\mathbf{a}^\star$ holds the coefficients of $\pi_{\mathrm{EI}}^\star$. A correction learned where the model-based ~~action~~ correction is large is ~~reduced automatically when the local MB solution requires little intervention~~therefore reduced where the local model-based correction is small, rather than being transferred as a fixed altitude offset.

Generalization: The final block evaluates whether the policy structure learned at the nominal condition remains useful when the operating condition changes. Two off-nominal conditions are ~~considered:~~ tested, $(z^*/R,\, v_x^*/v_h) = (2.5, 0.045)$ and $(1, 0.113)$. ~~For eachcondition, a~~ At each, the baseline policy π_{BL} is flown, and a model-based policy π_{MB} ~~are~~ is computed with the frozen ~~identified~~ DT. The best nominal RT correction is then transferred in the coordinates of the condition-specific model-based reference correction and denoted by $\pi_{\mathrm{EI}}^{\mathrm{tr}}$. Where an additional condition-specific EI search is performed, its best physical policy is denoted by $\pi_{\mathrm{EI}}^\star$. These tests ~~distinguish~~ separate direct transfer of the nominal correction from a local ~~, proof-of-concept extension of the~~ RT search at a new operating condition~~; they do not constitute convergence of a surrogate over the full velocity-clearance domain~~.

III. Results and Discussion

A. Baseline Response

~~The first experimental case employs only the cascaded PID controller available in the original Crazyflie firmware.The most direct target for this baseline crossing is a constant-clearance trajectory at $(z^*/R,\, v_x^*/v_h) = (1.5, 0.079)$. The desired trajectory is therefore the straight horizontal line denoted by~~ Fig. 3 shows the baseline crossing at the nominal condition, flown with the onboard cascaded PID controller alone. The dashed line is the constant reference π_{BL} ~~and reported with a dashed line in Fig. 3. This case establishes the performance baseline because the controller acts exclusively~~ . The controller acts only through feedback, without ~~any anticipation of the approaching~~ anticipating the aerodynamic interaction.

The solid line in Fig. 3 reports the mean experimental trajectory $\mathrm{EXP}_{\mathrm{BL}}$ obtained from ~~five~~ three repetitions, while the surrounding shaded region represents the corresponding $\pm$RMS run-to-run dispersion. ~~At each streamwise position~~After the repetitions are aligned at the leading-edge crossing and interpolated on the samples of the first one, this quantity is ~~defined as~~ $\mathrm{RMS}(x) = [N_r^{-1}\sum_{j=1}^{N_r}(z_j(x) - \bar{z}(x))^2]^{1/2}$, with ~~$N_r = 5$~~ $N_r = 3$. The band therefore describes repeatability about the measured mean and is not an uncertainty interval for that mean. Before ~~reaching~~ the obstacle, the vehicle ~~maintains an almost~~ holds a nearly constant clearance close to the commanded value~~, supporting satisfactory tracking at this modest forward speed away from the surface interaction. The response changes markedly over the obstacle: the vehicle initially rises to approximately~~ . Over the obstacle, it first rises to about $z/R = 2.5$ and ~~subsequently~~ then settles around $z/R = 2$. The additional thrust generated in ground effect produces the initial overshoot. The feedback controller reacts by reducing the motor speeds and commanded thrust, but this corrective action is insufficient to recover $z^*/R = 1.5$ while the vehicle remains over the obstacle. The cascaded PID is tuned around hovering and contains no model that can anticipate the surface-induced thrust augmentation.

The opposite transient occurs at the trailing edge. Ground-effect augmentation disappears while the motor command is still reduced by the preceding feedback action, producing an undershoot down to ~~approximately $z/R = 0.5$~~about $z/R = 0.4$. The vehicle then ~~gradually~~ recovers the commanded clearance, ~~which is reached only about $40R$~~ to within $0.1R$, about $35R$ downstream of the ~~obstacle~~trailing edge. Both excursions ~~are substantially larger than~~ exceed the run-to-run dispersion, ~~indicating~~ so they reflect a repeatable interaction rather than experimental scatter. ~~The baseline experiment consequently provides the reference cost for all subsequent performance comparisons and the first factual record used during the information phase.~~

B. Information Phase

Each information-oriented ~~policy is constructed by fixing a set of longitudinal nodes and allowing the optimizer to modify the local vertical reference at those nodes. Interpolation between consecutive values produces the continuous reference $z_\pi(x)$ deployed through the onboard PID controller. The~~

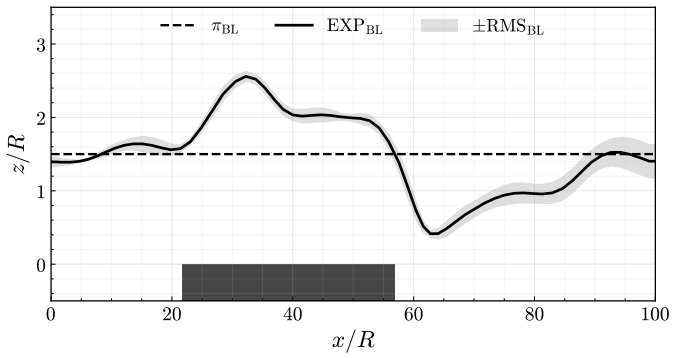


Fig. 3. Baseline policy π_{BL} and mean experimental response $\mathrm{EXP}_{\mathrm{BL}}$ over ~~five~~ three repetitions. The shaded band represents the corresponding $\pm$RMS dispersion.

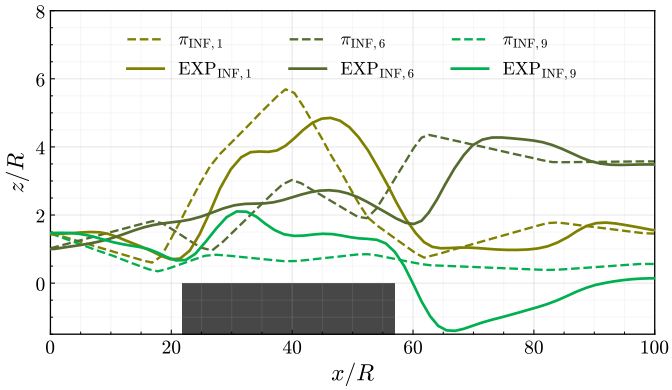


Fig. 4. Representative information-oriented policies and physical trajectories in the broad and ~~near-wall~~ low-clearance domains.

~~longitudinal locations remain fixed , while the optimized nodal heights determine which clearances and altitude variations are requested during the crossing. The bounds of the design domain apply to~~ reference is defined by its values at fixed streamwise nodes and interpolated between them. The design domain bounds this reference, not ~~to the DT or physical vehicle~~ the trajectory, which ~~can depart~~ departs from it under the closed-loop dynamics.

~~Two consecutive design domains are used.~~ The first six policies are ~~generated~~ designed in the *broad domain*, ~~with the reference bounded within $0.5 < z/R < 6$, allowing the optimizer to explore a broad range of vertical excitations. After these six policies, the upper bound is reduced to $z/R = 1.5$ for the remaining six episodes, defining the *near-wall domain*. The purpose of this restriction is to add information near the wall~~$0.5 \le z/R \le 6$, and the following ones in the *low-clearance domain*, $0.3 \le z/R \le 1.5$, where the ground-effect response is ~~expected to be~~ strongest and most sensitive to A and p. ~~Representative references and physical responses from both domains are reported in~~ Fig. 4 shows representative references and trajectories. The broad-domain policies $\pi_{\mathrm{INF},1}$ and $\pi_{\mathrm{INF},6}$ request ~~markedly different altitude~~ different clearance histories across the obstacle, whereas the ~~near-wall~~ low-clearance policy $\pi_{\mathrm{INF},9}$ ~~concentrates~~ keeps the crossing at smaller clearances. ~~The physical curves indicate the clearances actually traversed by the vehicle and are the trajectories used during~~ These trajectories are the ones used in factual replay and identification.

The estimates of the model parameters $\boldsymbol{\theta}_p = [A, p]^{\mathsf{T}}$ are reported in Fig. 5(a). Both coefficients ~~are initialized with the Cheeseman-Bennett values , $A = 1/16$~~ start from the Cheeseman–Bennett values and ~~$p = 2$, and change substantially~~ change to $A = 0.104$ and $p = 1.34$ after the first ~~adjoint identification, performed using~~ identification, which uses only the baseline record ~~in the gray region~~(gray region). Each subsequent point is obtained after appending one new episode to the replay buffer and repeating the cumulative identification. ~~The episodes designed with references in the broad domain $0.5 < z/R < 6$ occupy the olive region , while those using the near-wall domain $0.5 < z/R < 1.5$ occupy~~ Broad-domain episodes lie in the olive region and low-clearance episodes in the green region.

The E-optimal quantity $\lambda_{\min}(\mathbf{F})$ is reported in Fig. 5(b). It is the cumulative, locally scaled Fisher-information proxy available along the weakest joint parameter direction and is compared for the designed sequence, denoted by *Designed* π_{INF}, and the counterfactual sequence *Repeated* π_{BL}. The latter ~~repeats the same~~ counts the measured baseline episode ~~at 0.35 m/s, including~~ once per cumulative episode, with its sampling and residual level, ~~for the same number of cumulative interactions~~so that it measures repeated-design information under independent noise rather than new experiments. At the first information-oriented mission, corresponding to cumulative episode 2, $\lambda_{\min}(\mathbf{F})$ reaches 0.676 for *Designed* π_{INF} and 0.060 for *Repeated* π_{BL}. ~~The designed episode therefore provides more than one order of magnitude more information along the weakest~~ Under the adopted local scaling, the design score of the designed episode is therefore more than ten times larger along the least-observed direction.

~~Across the following~~ Over the next five broad-domain episodes, $\lambda_{\min}(\mathbf{F})$ ~~continues to increase, but with a much smaller slope, reaching~~ rises only from 0.676 to 1.000~~after the sixth information-oriented policy~~. This value is dimensionless and set by the scaling of Section II-C; it is not a normalization. The GE-parameter estimates also continue to move and reach $A = 0.218$ and $p = 2.792$ at the end of this stage. Their variation is smaller than after the initial identification, although they are not yet stable. The ~~estimated parameter correlation starts at $\rho_{A,p} = 0.974$~~ correlation $\rho_{A,p}$ of A and p, computed from $\boldsymbol{\Sigma}_\theta$, drops from 0.974 for the baseline record ~~and varies non-monotonically as new trajectories are added, reaching 0.883 after the six~~ to 0.783 after the first informative episode. It then rises to 0.883 over the remaining broad-domain ~~policies. The reduction supports the conclusion that the designed missions add a new sensitivity direction, but the remaining value is still high~~episodes, as further broad-domain data mainly reinforce the same coupled direction.

The broad domain nevertheless provides a useful first identification of the ground-effect law, especially ~~away from the wall~~at larger clearance. The corresponding function, denoted by $f_{\mathrm{GE}}(0.5 < z/R < 6)$, is compared with the ~~Cheeseman-Bennett~~ Cheeseman–Bennett initialization and the final estimate in Fig. 6(a). The local linearized uncertainty

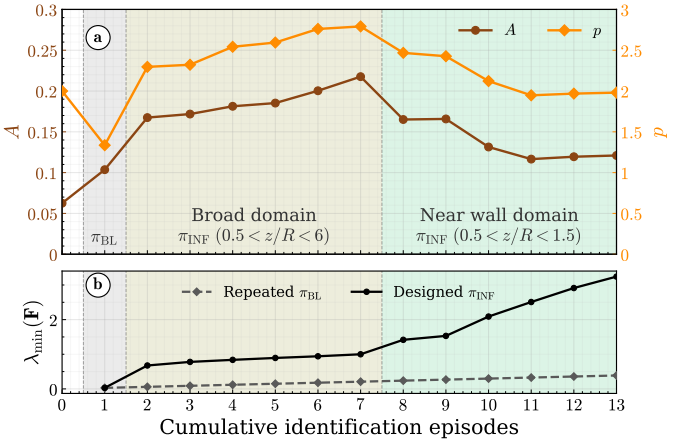


Fig. 5. Sequential identification of A and p and cumulative E-optimal information obtained with the designed policies and the repeated-baseline counterfactual.

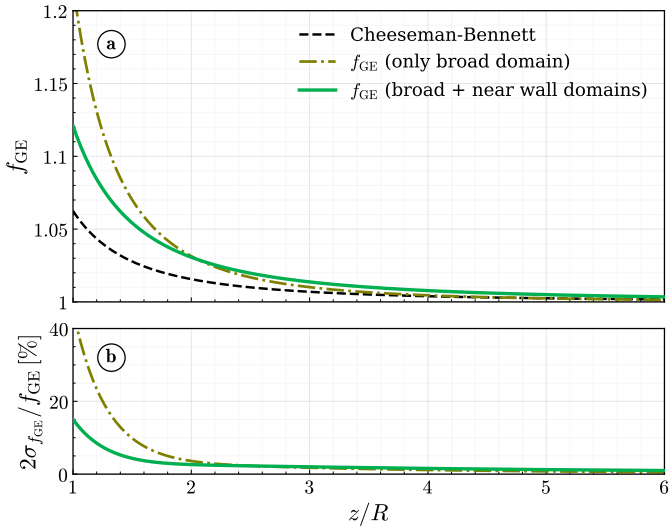


Fig. 6. Identified ground-effect function and propagated uncertainty after the six broad-domain experiments and after combining the broad and ~~near-wall~~ low-clearance domains.

indicator $2\sigma_{f_{\mathrm{GE}}}/f_{\mathrm{GE}}$ is reported in Fig. 6(b). ~~It describes uncertainty in the total thrust factor rather than in the smaller augmentation $f_{\mathrm{GE}} - 1$, and it is not an exact confidence interval.~~ The olive dash-dot curve predicts a stronger effect than ~~Cheeseman-Bennett~~ Cheeseman–Bennett at small clearance, but ~~this portion of the curve is still associated with substantial local uncertainty. Above approximately~~ there the indicator is still 45% at $z/R = 1$. Above about $z/R = 2$, ~~the indicator is already small~~ it is below 4% because the broad trajectories provide coverage there and the total factor approaches unity.

The ~~observed clearance coverage and the uncertainty remaining below $z/R = 2$ motivate the restriction of the reference domain to $0.5 < z/R < 1.5$. The earlier records remain in the cumulative replay buffer; only the admissible references for the next designs are changed. This second~~ low-clearance stage produces trajectories such as $\pi_{\mathrm{INF},9}$ in Fig. 4, which spend more of the crossing in the region where f_{GE} is most sensitive to the two coefficients.

The effect of this decision is visible in both panels of Fig. 5. The slope of $\lambda_{\min}(\mathbf{F})$ increases again when the ~~near-wall~~ low-clearance episodes are introduced, and its cumulative value rises from 1.000 to 3.242~~. Twelve repetitions of the baseline would instead~~, of which 0.733 comes from the two archived records. Repeating the baseline record over the same 13 cumulative episodes would reach only 0.387. At the same time, A and p shift toward smaller values and stabilize from approximately cumulative episode 11 onward. The final estimates are $A = 0.121$ and $p = 1.980$.

The ~~parameter correlation decreases further~~ correlation then decreases to $\rho_{A,p} = 0.818$, ~~although it remains appreciable~~which is still high. This residual coupling follows from the ~~structure of the selected~~ power law and ~~the range~~ from the limited range of clearance that can be ~~explored safely: coordinated~~ flown. The sensitivities $\partial f_{\mathrm{GE}}/\partial A = (R/z)^p$ and $\partial f_{\mathrm{GE}}/\partial p = A(R/z)^p \ln(R/z)$ are nearly proportional over a narrow range of clearance, and the second vanishes at $z = R$ and changes sign across it. Coordinated variations of A and p ~~can generate very similar~~ therefore produce nearly identical $f_{\mathrm{GE}}(z/R)$ curves. Additional data can shrink the set of plausible coefficient pairs without removing this intrinsic elongated direction completely. The E-optimal criterion has therefore increased the information along the weakest observable direction, while the two physical parameters remain partially coupled.

Returning to Fig. 6, the final model ~~changes only moderately at large clearance but differs substantially from~~ is close to the broad-domain estimate ~~below~~ above $z/R = 2$ ~~. The most important improvement in this regionis the contraction of the relative uncertainty. The coefficient stabilization, the limited further reduction of $\rho_{A,p}$, and the improved confidence in the control-relevant part of f_{GE} are taken together as the stopping condition for~~ and differs from it below, with $f_{\mathrm{GE}} = 1.121$ instead of 1.218 at $z/R = 1$. In this region, the indicator $2\sigma_{f_{\mathrm{GE}}}/f_{\mathrm{GE}}$ falls from 45% to 15% at $z/R = 1$ and from 9.9% to 4.3% at $z/R = 1.5$. These observations and the replay of Fig. 7 are the basis for ending the information phase (Section II-A). Obtaining additional information would require trajectories even closer to the obstacle, reducing the safety margin and increasing the likelihood of contact or hardware damage. The accessible clearance is also bounded by the physical vertical envelope of the vehicle, which prevents meaningful exploration arbitrarily close to the surface.

The final ~~coefficients differ substantially from the Cheeseman-Bennett initialization: the exponentreturns close to two, but the identified amplitude~~exponent, $p = 1.980$, is close to the Cheeseman–Bennett value $p = 2$, whereas the amplitude, $A = 0.121$, is almost twice ~~the classical value~~1/16. A ground effect stronger than the classical prediction has also been measured on a tethered Crazyflie 2.0 [3] and on a small multirotor [61]. This supports ~~the value of~~ a system-specific information phase when a low-order model is transferred from an isolated hovering rotor to a complete Crazyflie crossing a finite obstacle. ~~It does not imply that the two-parameter law resolves every transient feature; it establishes a more~~

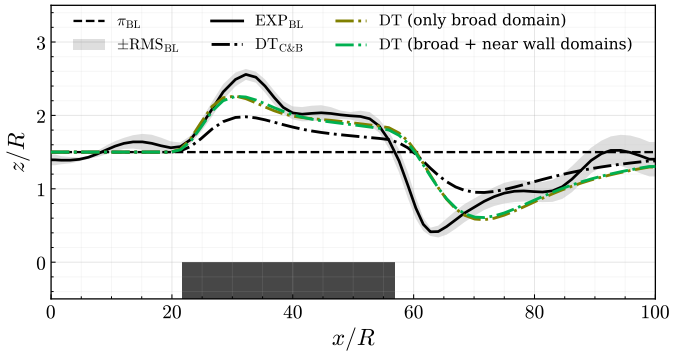


Fig. 7. Baseline experiment replayed with the classical initialization, the DT identified using only the broad domain, and the final DT identified using the broad and ~~near-wall~~ low-clearance domains.

~~representative static model over the range that can be tested safely.~~

The relevance of this improvement to control is assessed through the factual replay reported in Fig. 7. The baseline experiment is simulated using the ~~Cheeseman-Bennett~~ Cheeseman–Bennett coefficients, the coefficients identified using only the broad domain, and the final coefficients identified by combining the broad and ~~near-wall domains. The model initialized with Cheeseman-Bennett predicts the qualitative trend: the increased thrust raises the vehicle above the obstacle, and the removal of that contribution produces a~~ low-clearance domains. With the Cheeseman–Bennett coefficients, the DT predicts the rise over the obstacle and the loss of altitude downstream~~. Its magnitude is nevertheless underestimated~~, but underestimates both. Both identified models predict a larger rise over the obstacle and a stronger post-obstacle descent, producing a closer match with the experiment.

The broad-domain and final replays ~~are already very similar for this baseline case. The near-wall episodes are still valuable because they improve the confidence and the local form of f_{GE} where the interaction is strongest, but they produce only a small additional change in the replay~~ of π_{BL} ~~. This result supplies the control-oriented stopping argument: the~~ are nearly identical. The model obtained after the broad domain therefore already reproduces the dominant baseline response, and the ~~combined dataset refines the uncertain near-wall portion without materially changing the DT prediction for the baseline mission~~low-clearance episodes refine f_{GE} below $z/R = 2$ without changing this replay.

~~Overall, the information phase replaces a generic ground-effect approximation with a model specialized to the tested vehicle and operating envelope. The broad-domain policies establish the global response, while the near-wall domain adds information and reduces uncertainty close to the wall. Further refinement would demand experiments with a smaller physical safety margin, whereas the objective of RT also requires the identified DT to be used for control improvement. The combination of stable coefficients, limited remaining correlation improvement, reduced near-wall uncertainty, and nearly unchanged baseline replay between the intermediate and final models therefore provides a practical and transparent justification for ending identification and proceeding to the model-based policy~~The information phase thus replaces the generic Cheeseman–Bennett law with a law identified for the tested vehicle and clearance range.

C. Model-Based Policy

~~Once the information phase is completed, the identified GE parameters are frozen and the resulting DT is used to compute~~ Fig. 8 compares the baseline policy with the model-based policy π_{MB}~~. The objective is to exploit the DT, now augmented with the final identified ground-effect model, to construct an anticipatory vertical reference for the onboard cascaded PID controller. The reference is parameterized on a dense spatial grid with nodes separated by $\Delta x \simeq R$ and linearly interpolated between adjacent nodes. Their vertical coordinates are optimized by minimizing the tracking cost introduced in Section II-C, with respect to the nominal clearance $z^*/R = 1.5$ and subject to the prescribed safety constraints.~~

~~The comparison with the baseline policy is reported in Fig. 8. The~~, obtained by minimizing J_{MB} of Eq. (39). Unlike the constant reference π_{BL}, ~~executed through the built-in PID controller, can react to the aerodynamic interaction only after a tracking error has developed. By contrast,~~ π_{MB} begins to reduce the commanded clearance before the leading edge and over the first part of the obstacle. This anticipates the thrust augmentation produced by ground effect. The reference is subsequently raised around the trailing edge, prompting the feedback controller to increase thrust before the aerodynamic contribution is removed. Farther downstream, the correction gradually returns to zero as the vehicle recovers the nominal clearance.

These anticipatory actions ~~produce a markedly lower experimental~~ lower the measured trajectory over the obstacle~~than the baseline response. The dominant overshoot~~is ~~attenuated and~~, attenuate the overshoot, and shorten the downstream recovery~~is faster. A substantial~~. An undershoot nevertheless remains after the trailing edge, ~~with a local minimum comparable to that obtained~~ to $z/R \approx 0.5$ against 0.4 under π_{BL}. The improvement therefore comes mainly from reducing the excursion over the obstacle and shortening the downstream tracking transient, rather than eliminating every local extremum.

~~The performance change can be quantified with the same reward definition used throughout the subsequent policy search.The~~ With the reward of Eq. (38), the DT reward increases from ~~approximately $\mathcal{R}_\circ = 0.68$~~ $\mathcal{R}_\circ = 0.676$ for π_{BL} to ~~$\mathcal{R}_\circ = 0.97$~~ 0.968 for π_{MB}~~. The corresponding physical reward increases from approximately $\mathcal{R}_\bullet = 0.54$ to $\mathcal{R}_\bullet = 0.85$. Their consistent ordering supports the conclusion that the identified DT contains sufficient causal information~~, and the physical reward from $\mathcal{R}_\bullet = 0.544$ to 0.855. Both rewards rank π_{MB} above π_{BL}, so the DT is accurate enough to design a policy that improves physical tracking, ~~while the difference in their absolute values quantifies~~ and the gap between the two rewards measures the remaining transfer error.

~~This difference is also visible in the trajectory replay~~The DT prediction DT_{MB} in Fig. 8 shows the same gap. Over the ob-

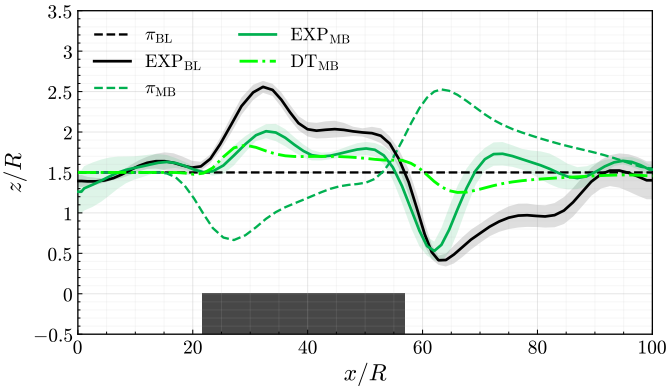


Fig. 8. Experimental baseline and model-based policies, together with the DT prediction of π_{MB}.

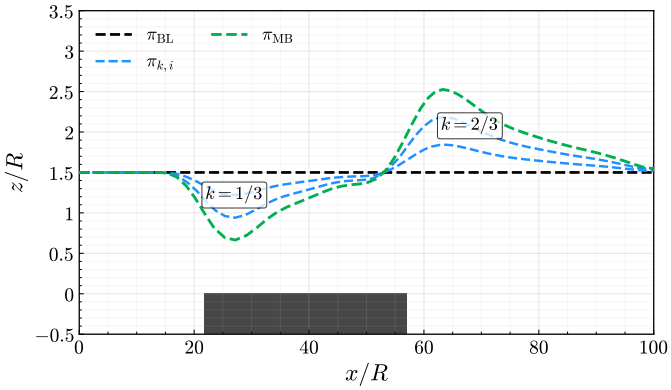


Fig. 9. Intermediate reference policies $\pi_{\lambda,i}$ between the baseline and model-based policies.

stacle, $\mathrm{DT}_{\mathrm{MB}}$ and $\mathrm{EXP}_{\mathrm{MB}}$ follow similar trends, although the virtual response still underestimates the measured overshoot. The ~~larger discrepancy occurs around~~ largest difference occurs at the trailing edge~~: the~~. The DT predicts that ~~the reference increase should almost suppress~~ raising the reference almost suppresses the downstream undershoot, whereas ~~a pronounced loss of clearance remains experimentally~~the vehicle still descends to $z/R \approx 0.5$. The identification loss covers only the samples above the obstacle, so the DT is not fitted to this post-edge response. The DT therefore predicts the correct spatial structure and benefit of the compensation but overestimates its effectiveness in the transient regions where the static GE model is least representative.

~~The model-based policy consequently provides both a performance improvement and a well-informed reference for the next phase. A global model-free search would discard the safety and data-efficiency benefits already supplied by the identified DT. Instead, the performance phase explores the neighborhood of π_{MB} and learns the GP discrepancy between the DT and physical rewards. The reward-transfer surrogate retains the useful ranking supplied by the DT while correcting the residual mismatch that prevents the best DT policy from coinciding with the best physical policy.~~

D. Performance Phase

~~Once the model-based policy has been obtained, the GE parameters are frozen and the campaign proceeds to the performance phase. The purpose of this phase is to learn~~ The performance phase learns the GP discrepancy between ~~the~~ DT and physical rewards. ~~The comparison discussed in~~ Section III-C ~~revealed~~ showed that the DT ~~remains optimistic , particularly around the leading and trailing edges of the obstacle~~is optimistic around the trailing edge, where the static ~~GE correction~~ ground-effect law cannot represent the complete transient ~~aerodynamic response. Rather than increasing the complexity of the physical model, RT~~ response. The reward-transfer surrogate accounts for the control-relevant part of this ~~structural error through the reward-transfer surrogate. The resulting search retains the safety and data efficiency of the model-based policy while allowing the best physical policy to depart from the best DT policy.~~ error without enlarging the model.

~~Only performance-oriented policies are considered at this stage. The information-oriented policies used during identification are excluded because their large altitude excursions were designed to excite the modelparameters, not to maintain the nominal clearance $z^*/R = 1.5$.~~ The initial reward-transfer surrogate dataset ~~therefore~~ contains the two performance-oriented policies already available, π_{BL} and π_{MB}, for which both DT and physical rewards have been evaluated. ~~Two additional policies are then introduced by applying one and two thirds of the model-based reference correction to the baseline reference. The resulting policies $\pi_{\lambda,i}$, with $\lambda \in \{1/3, 2/3\}$, are reported in~~ Fig. 9 ~~. Since they lie on the direct interpolation path between the two existing policies , they are expected to produce intermediate rewards. Their main role is consequently to establish the global DT-to-physical reward trend~~ shows the two interpolation policies $\pi_{\lambda,i}$ of Section II-C, which apply one and two thirds of the model-based correction and calibrate the affine ~~component of the reward-transfer surrogatebefore the GP discrepancy is introduced~~part of the surrogate.

The calibration ~~must subsequently be refined near the region of greatest practical interest, namely the model-based policy. Exploring around~~ is then refined near π_{MB}~~tests whether small changes to an already safe and effective policy can improve the physical response. This local strategy avoids the experimental cost and risk of an unconstrained model-free search. It also keeps the policy parameterization deliberately compact: a high-dimensional~~ . Three coefficients keep the number of flights needed to train the GP small; a richer correction could describe more ~~complex shapes , but would require substantially more physical tests before the Gaussian process could resolve their effect on performance~~shapes but would need more flights.

~~In the present work, local corrections are generated from three Gaussian basis functions centered on the model-based reference at $x/R \simeq 35$, 54, and 70. The first two centers , denoted by~~ The basis centers A and B ~~,~~ lie above the obstacle ~~, whereas C acts on the downstream recovery.Each basis has one bounded amplitude, so that $\mathbf{a} = [a_A, a_B, a_C]^\top$~~

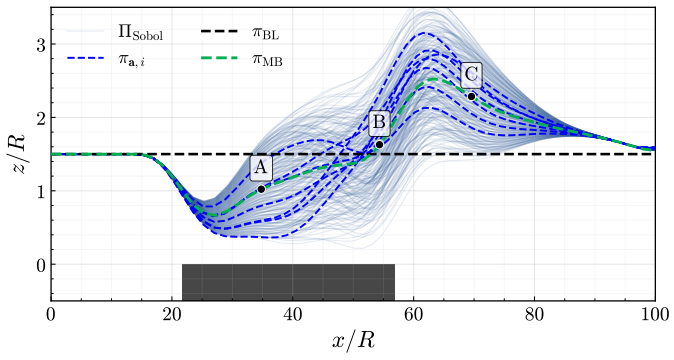


Fig. 10. ~~Safe~~ Representative unscreened samples of the Sobol ~~policy~~ library ($|a_j| \le 0.8$), basis locations, and ~~experimentally tested local~~ the seven Sobol policies $\pi_{\mathbf{a},i}$ flown.

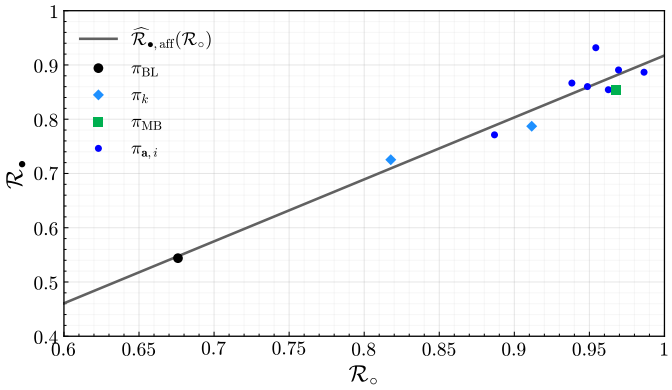


Fig. 11. Reward-transfer surrogate after local policy exploration.

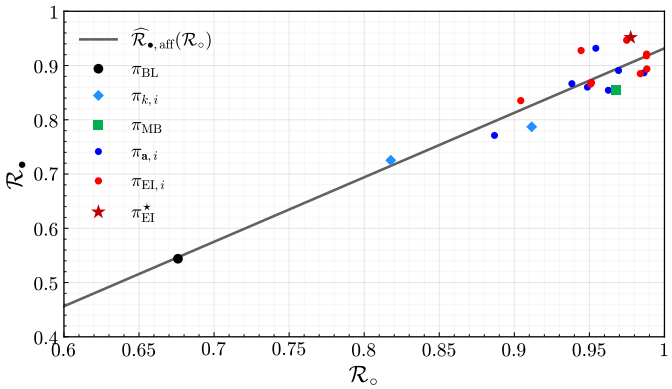


Fig. 12. Reward-transfer surrogate after the EI-selected physical evaluations; the star denotes the best policy $\pi^\star_{\text{EI}}$.

~~completely specifies the local departure from π_{MB}, with $a_A, a_B, a_C \in [-1, 1]$. A space-filling Sobol sequence is used to generate the faded-blue candidate family reported in~~ and C downstream of it (Fig. 10). Fig. 10 ~~. The~~ shows representative samples of the Sobol library (faded blue) and the seven policies flown (dark-blue ~~dashed references in the same figure identify the small subset selected for physical testing. The subset spans different amplitude combinations and deliberately includes both apparently unfavorablechanges such as increasing~~ dashed). They combine different amplitudes and include changes that look unfavorable, such as raising the reference above the obstacle or ~~reducing it downstream and corrections~~ lowering it downstream, as well as changes that extend the ~~compensating action already suggested by~~ correction of π_{MB}. This spread is required to teach the surrogate which local shape changes improve the experiment and which merely appear favorable in the DT.

The reward pairs obtained after this local exploration are reported in Fig. 11. As expected, the two interpolated policies lie approximately along the trend joining π_{BL} and π_{MB}. Most ~~of the tested Sobol policies instead occupy a narrow neighborhood~~ Sobol policies have DT rewards close to that of π_{MB} ~~on the DT-reward axis, while~~ , whereas their physical rewards ~~remain sufficiently separated to reveal~~ range from 0.771 to 0.932, which reveals a policy-dependent GP discrepancy. These measurements ~~improve the local calibration of $\widehat{\delta}(\mathbf{a})$: the affine relation continues to describe the overall relation between the DT and physical rewards, whereas the Gaussian process learns how policies with similar DT performance can be ranked differently by the physical environment~~calibrate the GP discrepancy near π_{MB}. Once measurements exist on different sides of the model-based policy, the GP interpolation uncertainty can guide the next experiment through Expected Improvement. ~~The reward markers are computed from the cleaned mean trajectories; run-to-run dispersion is assessed in the trajectory comparisons and is not represented as a statistical error bar in the surrogate plots.~~

The EI-selected policies are superposed on the calibrated reward-transfer surrogate in Fig. 12. ~~The acquisition~~ EI does not seek ~~an immediate~~ a reward increase at every iteration~~: some~~ . Some policies are selected ~~because their GP interpolation uncertaintymakes them informative candidates~~for their GP uncertainty, and their physical reward can ~~therefore~~ remain below the ~~incumbent. Nevertheless, after only a few EI-guided evaluations, the search identifies~~ best reward $\mathcal{R}^+_\bullet$ observed so far. The ninth of the ten EI-selected policies is $\pi^\star_{\text{EI}}$. Relative to π_{MB}, its DT reward increases only from ~~approximately~~ 0.968 to 0.978, whereas the physical reward rises from ~~approximately~~ 0.855 to 0.952. The ~~marked difference between these two increments is the improvement enabled by the learned GP discrepancy: the DT alone regards~~ DT alone ranks the two policies as nearly equivalent, ~~while the physical observations reveal a substantial performance gain. Other high-reward evaluations approach a similar reference shape, supporting a repeatable control strategy rather than proving convergence to a unique optimum~~and the physical rewards, through the GP discrepancy, reveal the difference. The fifth EI policy, whose coefficients differ from those of $\pi^\star_{\text{EI}}$ by at most 0.024, reached 0.947, so the gain is not tied to a single policy. The coefficients of $\pi^\star_{\text{EI}}$ lie at least 0.30 from the faces of the box $[-1, 1]^3$, which therefore does not bind at the best policy. The best Sobol policy reached 0.932.

The physical meaning of this reward increase is reported in Fig. 13, where the baseline, model-based, and best EI policies are compared with their mean experimental responses. Above the obstacle, $\pi^\star_{\text{EI}}$ maintains a lower reference over a longer streamwise interval than π_{MB}, which ~~slightly~~ reduces

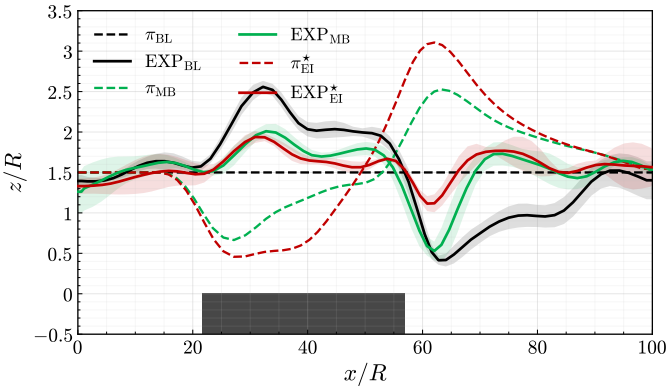


Fig. 13. Desired and mean experimental trajectories for π_{BL}, π_{MB}, and $\pi^{\star}_{\mathrm{EI}}$.

the remaining positive ~~tracking excursion. The command then rises earlier and more rapidly near~~ excursion. Near the trailing edge~~. The vehicle consequently reaches a somewhat larger clearance before leaving the obstacle and is better prepared for the subsequent loss of aerodynamic support~~, the command then rises higher, to about $z/R = 3.1$ against 2.5 for π_{MB}. The downstream ~~undershoot is markedly reduced~~ minimum rises from $z/R \approx 0.5$ to 1.1, and the recovery toward $z^*/R = 1.5$ ~~becomes~~ is smoother. This ~~reduction of the~~ smaller post-obstacle excursion ~~supplies the dominant reward gain and clearly~~ separates $\pi^{\star}_{\mathrm{EI}}$ from ~~the purely model-based policy~~π_{MB}; the two predictive solutions remain more similar over the obstacle itself.

The same comparison is reconstructed from video frames in Fig. 14. Whereas the curves in Fig. 13 are ~~obtained by averaging five~~ averages of three repetitions of each case, each row of Fig. 14 follows one representative flight. The single-run paths ~~reproduce the ordering observed in~~ show the same ordering as the mean trajectories~~, supporting the repeatability of the measured improvement. In particular, the three paths superposed in Fig. 14(c) make the large departure of the baseline policy immediately visible~~. The model-based and EI responses remain closer above the obstacle, while the best EI policy preserves the nominal clearance more effectively after the trailing edge.

E. Generalization Across Operating Conditions

~~Up to this point, the reward-transfer surrogate has been calibrated at the single nominal operating condition $(z^*/R,\, v_x^*/v_h) = (1.5, 0.079)$. At this condition , the model-based policy produced a clear improvement over the baseline policy, and only a few additional physical evaluations were required for the GP discrepancy δ to capture part of the residual mismatch between the DT and physical rewards. Expected Improvement could consequently identify a policy with substantially better physical performance even though its DT reward was only marginally higher than that of π_{MB}. The final campaign block investigates whether this policy correction~~ The final stage tests whether the correction learned at the nominal condition remains useful when both the desired clearance and the forward ~~velocity are changed~~speed change. ~~Two additional operating conditions are considered: a low-speed~~Fig. 15 shows the lower-speed, larger-clearance crossing at $(z^*/R,\, v_x^*/v_h) = (2.5, 0.045)$ and ~~a~~ the higher-speed, ~~small-clearance~~ smaller-clearance crossing at $(1, 0.113)$. The ~~trajectory comparisons are reported in Fig. 15. At each condition, the built-in PID controller executes the condition-specific baseline policy π_{BL}, while a new model-based policy is computed using the same frozen parameter vector $\widehat{\boldsymbol{\theta}}_p$ identified at the end of the information phase. The~~ nominal best correction, $\mathbf{a}^{\star} = [-0.66, 0.70, 0.20]^{\mathsf{T}}$, is then transferred ~~by applying the same three coefficients $\mathbf{a} = [a_A, a_B, a_C]^{\mathsf{T}}$ relative to the condition-specific model-based policy. This scaling preserves the learned correction structure without imposing the nominal reference correction as a fixed altitude offset~~with the scaling of Section II-C, with $\mathbf{s} = [0.41, 0.53, 0.43]^{\mathsf{T}}$ at $(2.5, 0.045)$ and $[1.85, 1.51, 1.60]^{\mathsf{T}}$ at $(1, 0.113)$.

In both operating conditions, the condition-specific model-based policy markedly improves tracking relative to the corresponding baseline policy. At $(z^*/R,\, v_x^*/v_h) = (2.5, 0.045)$, the transferred policy provides a further, albeit modest, improvement. At the higher-speed, small-clearance condition $(1, 0.113)$, however, its physical reward is slightly lower than that of the model-based policy. ~~This loss is important: the~~ The transferred correction was learned from discrepancies measured only at the nominal condition~~$(1.5, 0.079)$, whereas the relative importance of the unmodeled transients changes with velocity, clearance~~. At $(1, 0.113)$, the vehicle spends 1.6 s over the obstacle instead of 2.3 s, the transferred coefficients leave the box $[-1, 1]^3$, and the ~~magnitude of the model-based action. Transfer therefore provides an informed initialization, but it does not replace condition-specific learning when the operating condition moves away from the region represented in the reward-transfer surrogate dataset~~policies operate at clearances where f_{GE} is least constrained. The discrepancy can therefore differ from the nominal one.

~~A direct comparison of rewards across these missions requires the different baseline difficulty to be removed. The DT and physical quantities are therefore normalized independently by their condition-specific baseline-policy values according to~~ Rewards at different conditions are compared through the scores of Eq. (45)~~. The DT and physical scores are denoted by~~ , $\mathcal{S}_{\circ}$ for the DT and $\mathcal{S}_{\bullet}$ ~~, respectively. By construction, $\mathcal{S}_e = 0$ for the corresponding baseline policy and a larger value indicates a greater~~ for the experiment. Each is the fractional reduction of the tracking cost ~~. This normalization places missions with different velocities and clearances on the same improvement scale and defines the normalized reward-transfer surrogate used across operating conditions~~relative to the baseline policy at the same condition, for which it is zero.

In ~~this more general formulation~~the condition-aware surrogate, the GP discrepancy depends on both the policy and the operating condition [62],

$$\widehat{\mathcal{S}}_{\bullet} = \widehat{\mathcal{S}}_{\bullet,\mathrm{aff}}(\mathcal{S}_{\circ}) + \widehat{\delta}(\mathbf{a}, z^*/R, v_x^*/v_h)\,. \tag{46}$$

~~The vector $\mathbf{a}$ contains the three coefficients that modify the condition-specific~~ Here, $\widehat{\mathcal{S}}_{\bullet,\mathrm{aff}}$ is the affine fit of $\mathcal{S}_{\bullet}$

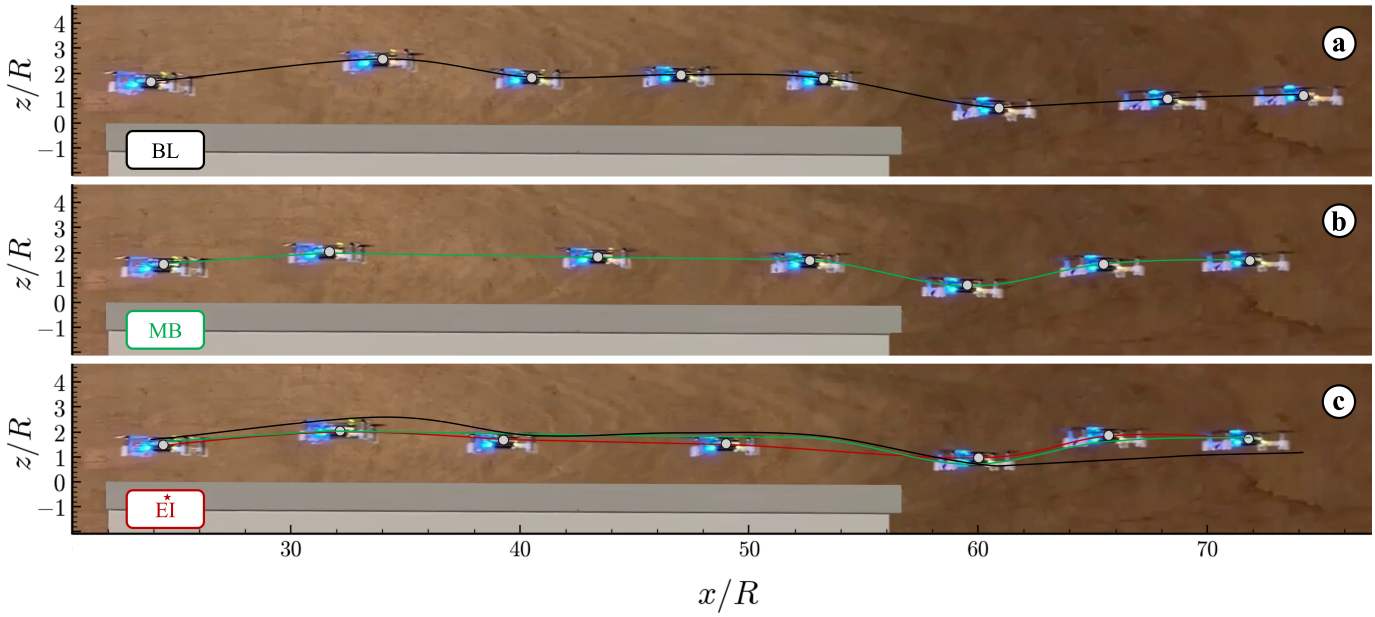


Fig. 14. Video-frame comparison of representative nominal-condition flights: (a) baseline policy π_{BL}, (b) model-based policy π_{MB}, and (c) best Expected-Improvement policy $\pi^{\star}_{\mathrm{EI}}$. The overlaid curves trace the measured vehicle path.

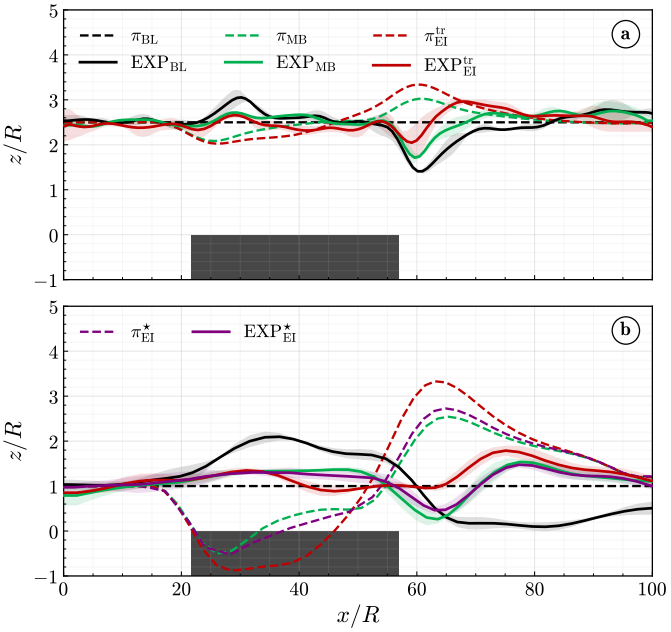


Fig. 15. Generalization at (a) $(z^*/R,\, v_x^*/v_h) = (2.5, 0.045)$ and (b) $(1, 0.113)$: PID, local model-based, transferred RT, and condition-specific EI policies.

on $\mathcal{S}_\circ$. As implemented, $\widehat{\delta}$ takes as standardized inputs the coefficients a_j divided by the local amplitude of the model-based ~~policy, while $(z^*/R,\, v_x^*/v_h)$ identifies the operating condition~~correction, the residual of their projection on the basis, v_x^*, and z^*/R. The scores are means over the flights of each policy. Their standard deviation over the flights, divided by the baseline cost and at least 0.01, weights the affine fit and enters the GP as noise. Additional experiments ~~distributed over the domain of interest would therefore train the normalized reward-transfer~~ over the velocity–clearance domain would train this surrogate to propose a condition-specific correction instead of ~~reproducing the same transferred policy everywhere. The present experiments provide an initial demonstration of this extension rather than a complete mapping of the velocity-clearance domain~~a single transferred one.

The normalized physical score is compared with its DT counterpart in Fig. 16. Black markers correspond to $(z^*/R,\, v_x^*/v_h) = (1.5, 0.079)$, orange markers to $(2.5, 0.045)$, and purple markers to $(1, 0.113)$. The transferred-policy diamond improves slightly on the model-based-policy square at the low-speed, larger-clearance condition. At the higher-speed, small-clearance condition, the ordering is reversed and the transferred policy performs slightly worse than π_{MB}. A correction calibrated at one operating condition therefore cannot be assumed to retain the same ranking elsewhere, which motivates extending the surrogate across the operating domain rather than relying on direct policy transfer alone.

The higher-speed, small-clearance condition is therefore used to demonstrate how the normalized reward-transfer surrogate can be extended locally. Two ~~EI-guided physical evaluations are added~~ EI rounds are run at $(z^*/R,\, v_x^*/v_h) = (1, 0.113)$; each proposes three distinct candidates, of which one is flown. The first explores an uncertain policy direction and performs poorly, yielding approximately $\mathcal{S}_\bullet = 0.57$, as indicated by the small purple marker in Fig. 16. Once this observation updates the condition-aware GP discrepancy, the second evaluation reaches a new condition-specific best of approximately $\mathcal{S}_\bullet = 0.83$, marked by the purple star, and exceeds the corresponding model-based policy. Its policy and physical trajectory are included in Fig. 15(b). Unlike the nominal transferred correction, this policy remains closer to π_{MB}, suggesting that the model-based policy was already near a favorable local optimum at the smaller clearance and higher

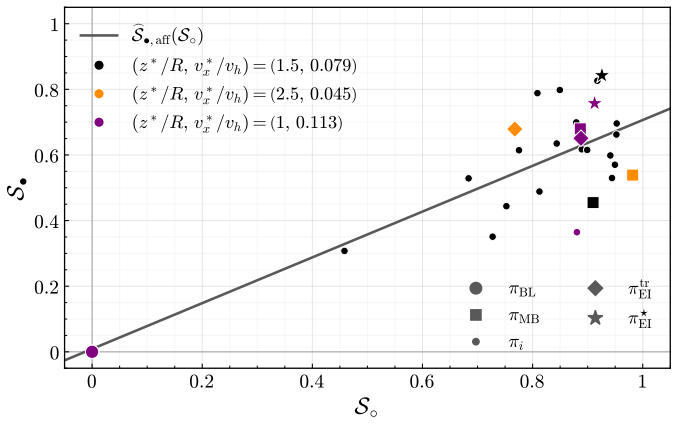


Fig. 16. Normalized reward-transfer surrogate across the nominal and two additional operating conditions.

velocity.

~~Only two condition-specific EI iterationsare used here, so the result is interpreted as a proof-of-concept local extension of the RT workflow rather than as convergence of a general~~ With only two EI iterations, this result is a proof of concept rather than a converged surrogate. At a new operating condition, the ~~local~~ frozen DT first supplies a model-based policy that ~~already improves substantially over~~ improves on the baseline policy. ~~When direct transfer fails to improve that policy, a small number of physical interactions can~~ A few physical episodes can then reveal how the GP discrepancy has changed and redirect the search toward a better condition-specific ~~solution. Repeating this process over the velocity-clearance domain would progressively calibrate $\widehat{\delta}(\mathbf{a}, z^*/R, v_x^*/v_h)$ and enable the generation of policies tailored to the requested operating condition while retaining the model-based policyas the safe and data-efficient reference~~policy.

IV. Conclusions and Outlook

This work demonstrated Reinforcement Twinning for ~~predictive~~ the trajectory control of a micro aerial vehicle crossing a finite obstacle in ground effect. ~~The information phaseprovided evidence that~~ In the information phase, the designed experiments ~~added more than visual diversity: the cumulative smallest Fisher eigenvalue grew much faster than for repeated baseline experiments, and restricting the design from the broad domain to the near-wall domain restored information gain as the former began to saturate. Although the identified model parameters remained correlated, the resulting ground-effect function was well constrained over the control-relevant clearance range. Parameter decorrelation was therefore not treated as the objective by itself; the relevant outcome was the improved identification of the weakest observable parameter direction and of the aerodynamic correction used by the controller.~~

~~Model-based optimization remained the backbone of the framework.~~ raised the smallest eigenvalue of the cumulative Fisher-information proxy to 3.242 over the baseline and twelve information-oriented episodes, against 0.387 for the baseline record repeated over the same episodes. The ~~identified DT generated an anticipatory~~ low-clearance records, two of them archived, gave most of this increase. The parameters remain correlated ($\rho_{A,R} = 0.818$), but the uncertainty indicator of f_{GE} fell to 15% at $z/R = 1$ and 4.3% at $z/R = 1.5$.

The DT produced a model-based policy that ~~improved tracking without replacing~~ raised the physical reward from 0.544 to 0.855, kept the onboard feedback controller~~and while satisfying the adopted virtual safety constraints. The remaining replay error then motivated the performance phase rather than invalidating the DT . Fractional~~ , and passed the DT clearance checks. Interpolation and Sobol policies then calibrated the reward-transfer surrogate, ~~while Expected Improvement directed subsequent physical experiments toward uncertain and potentially better policies . With few additional interactions, the resulting $\pi_{\mathrm{EI}}^{\star}$ outperformed the model-based policy by compensating for response features that the low-order static model could not rank accurately. In this way, the DT restricted the search to a screened, high-performing neighborhood, and physical rewards were used specifically to learn the GP discrepancy within that neighborhood~~and after nine EI-selected policies the search found a policy with physical reward 0.952, mainly by reducing the post-obstacle undershoot that the static model does not predict.

The off-nominal experiments clarified both the potential and the present limits of generalization. Directly transferring the nominal correction produced a modest gain at one operating condition but slightly underperformed the corresponding model-based policy at the other. After two condition-specific EI evaluations, the normalized reward-transfer surrogate identified an improved local policy. This result supports the data efficiency of local refinement, but it does not establish a surrogate that is converged over a continuous velocity-clearance domain.

The ~~conclusions are therefore limited to the tested envelope. The ground-effect model represents an unsteady rotor-surface interaction with~~ results are limited to a static two-parameter ~~law and cannot reproduce every transient feature. The physical experiments were constrained to $z/R \geq 0.5$, so the results do not support extrapolation toward contact, and generalization was examined at only two additional operating conditions.~~ ground-effect law, to the clearances flown (the lowest measured mean clearance over the obstacle was $0.68R$), and to two off-nominal conditions. Admissibility was checked only in the DT, which underpredicts the trailing-edge undershoot, so the DT screening reduces the risk of a policy without bounding it. Future work should ~~extend the campaign across a denser velocity-clearance domain, account explicitly for~~ cover a denser velocity–clearance domain, include repeated-measurement uncertainty in the ~~reward-transfer~~ surrogate, and ~~investigate richer~~ test unsteady aerodynamic models, moving landing surfaces, and onboard or real-time policy adaptation~~. Within the present limits, the campaign shows how RT can combine interpretable physics, virtual safety screening, and a targeted data-driven correction to improve control performance without requiring unconstrained model-free exploration~~.

ACKNOWLEDGMENT

The authors gratefully acknowledge the technical support of the von Karman Institute for Fluid Dynamics.

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